RESEARCH-ARTICLE

The Spread and Impact of Game Innovations: A Systematic Review of Academic and Applied Domains

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toxicity and extremism in online games

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The use of digital game technologies beyond entertainment has evolved from a niche practice to a well-established area of applied innovation [Sawyer & Rejeski, 2002; Future Market Insights, 2023]. Understanding these impacts is important for professionals in entertainment technology—researchers, teachers, students, developers, and policy makers—who, perhaps unknowingly, contribute to activities in other sectors. This systematic review investigates the use and impact of game technologies beyond entertainment to assess how the field contributes to other sectors through both its original innovations and those it has catalyzed and advanced. Across the publications analyzed, educational and applied games account for most documented cases, reflecting their prominence within academic discourse. We review academic research articles, book chapters, news coverage, and other published accounts to develop a broad picture of how game innovations spread—from first-order impacts, in which commercial entertainment games are leveraged toward non-entertainment ends in areas such as education, science visualization, and health; to second-order contributions, in which technologies and design innovations (e.g., game engines, leaderboards, occlusion-culling techniques) created for games are applied in non-game contexts; to tertiary contributions, where the games industry has catalyzed advances such as temporal anti-aliasing, scalable server architectures, and joystick design now used widely elsewhere. The review characterizes how video-game design elements and technologies have enhanced diverse fields, the relative degree of contribution across domains, and emergent themes. While the distribution of work mirrors the weighting of academic research toward educational and applied contexts, it also reveals broader technological and social impacts extending beyond those traditions. By focusing on published reports, this review moves beyond claims about potential to document the demonstrated, cross-sector contributions of game technologies outside entertainment.

CCS CONCEPTS • General literature • Surveys and overviews • Interactive learning environments

Additional Keywords and Phrases: Literature review, Video games, Game-based technologies, Health

1 INTRODUCTION

The use of digital game technologies beyond entertainment has become commonplace [Sawyer & Rejeski, 2002; Future Market Insights, 2023]. With showcases of games for impact occurring annually across the globe and dedicated academic journals, it now appears as if applied and educational uses of games have moved beyond the hype cycle [Linden & Fenn, 2003] of overinflated expectations and are now within the plateau of productivity, expanding into an $8.3B global market in 2023 [Future Market Insights, 2023]. As game technologies are deployed in areas that include emergency preparedness, construction, education, automotive design, military, and health care, knowledge from one sector may be disconnected from others.

This study investigates how video game technologies contribute to other fields, the sectors that games now impact, and the common successes, challenges, or themes to how gaming technologies are used across disparate domains. Understanding the impact and uses of game technologies beyond entertainment is particularly important for professionals in entertainment technology, including researchers, teachers, students, developers, and policy makers who, perhaps unknowingly, contribute to activities in other sectors. As one survey of university game programs shows [Higher Education Video Game Alliance, 2015], 44% of graduates found employment in sectors outside of games, including education, technology, and government and defense. As the job sector of entertainment games continues to experience waves of disruption [Schreier, 2021], the software industry beyond it offers shorter work hours and higher pay scales [U.S. Bureau of Labor

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Statistics, 2023]. Better understanding of where game technologies are employed beyond entertainment might help such stakeholders navigate career decisions.

The idea of games serving purposes beyond entertainment has deep historical roots. While Wilkinson (2016) traces the use of games and play for serious purposes as far back as Plato, the use of games and simulations in contemporary formal education can be traced back as early as the late 1950s (Gredler, 1996), although it was not until the early 1970s that they became part of the instructional design movement. Abt (1970) was the first to formalize the concept of "serious games" in educational and civic contexts, while Gordon's publication of the same year (1970) *Games for Growth* highlighted their unique capacity to teach systems to students while also aiding motivation, conceptual understanding, and problem-solving.

This systematic review investigates the use and impact of game technologies beyond entertainment to characterize and assess how the field contributes to sectors beyond its own through both its original innovations and those it has crucially catalyzed and advanced as part of its work. Across the publications analyzed, educational and applied games account for the majority of scholarly attention—a reflection of the focus of academic research rather than of the review's scope. Here, we review academic research articles and chapters, news articles, and published accounts of games' impact beyond entertainment to develop a broad understanding of the spread of game innovations, from "first order" impacts in which commercial entertainment games are leveraged toward other, non-entertainment ends in fields such as education, science visualization, and health to "second order" contributions in which technologies (such as game engines) and innovations (from leaderboards to occlusion culling techniques) invented in the context of games are applied in non-game fields, to "tertiary" contributions for which the games industry catalyzed or crucially advanced techniques (temporal anti-aliasing, scalable server architectures, joysticks) now commonly leveraged elsewhere. This review does not seek to characterize the economic impact of video games broadly as such reports already exist. Instead, it seeks to characterize and illustrate the ways in which video game design elements and technologies have enhanced far-flung fields, the relative degree of contribution in varying domain, and emergent themes. By focusing only on published reports, this review attempts to take a conservative approach yet go beyond mere accounts of the potentials of game technologies to instead capture and describe their documented positive contributions beyond entertainment alone.

2 METHODS

Attempting to capture and describe the total contributions of video games and game technologies across all possible domains is an ambitious enterprise given that target subject "video games and game-related technologies" can be operationalized in myriad ways and at myriad levels of granularity and the target domains in which to search are, by definition, unlimited. The research questions guiding this work are: (1) What are the elements of game-related technologies that have been used in other sectors, and (2) What, if any, positive impacts have they made in those areas?

2.1 Inclusion and Exclusion Criteria

We scoped the review to include documented uses of game technologies and methods—treating games both as products/systems and as technologies/platforms—but we only included evidence reported in academic journals, trade publications, government documents, book chapters, news articles, dissertations, books, and technical reports. This conservative criterion excluded undocumented claims and marketing copy, ensuring that all included cases had publicly citable documentation. In this way, our approach emphasizes documented applications and impacts rather than broad economic trend estimates. Evidence for industry-level effects (e.g., diffusion into AI pipelines or motion-picture VFX) is included when concretely documented, but high-level economic multipliers or anecdotal attributions without citable sources were out of scope. In keeping with our inclusion criteria, we coded technology/platform transfers (e.g., advances in graphics hardware, per-vertex/per-pixel pipelines, and later GPU-accelerated computation) when the transfer was explicitly documented in eligible sources; broader sectoral narratives without documentation were out of scope.

Only reports that made claims of video games or game-related technologies making a demonstrable impact on other domains were included in this review. Cross media franchises such as Pokémon [1996] were included only when evidence pertained to its interactive media elements. Reports related to analog (paper-based or table-top board) games rather than digital were considered beyond this study and excluded as were reports of the general size and scope of economic activity of video game related technologies. Speculative articles or reports touting the potential of games or game-related technologies were also excluded. The research team elected to constrain the corpus to English language publications for simplicity and feasibility.

2.2 Establishing the Data Corpus

Figure 1 (below) details the three steps taken to establish the data corpus used in this study.

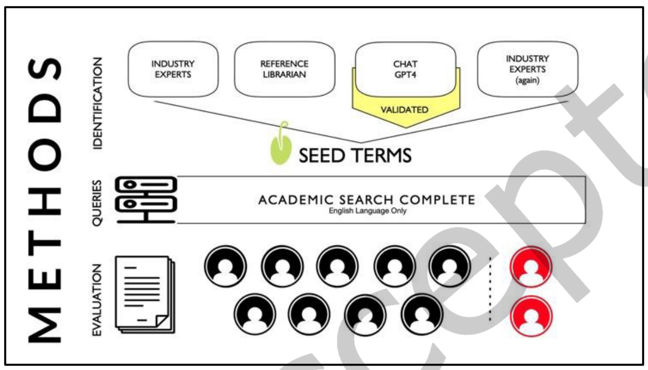

Figure 1: Overview of research methods used to construct the data corpus.

2.1.1 Identification

Our first task was to identify a set of game innovations known to have impacted other domains that could serve as “seed terms” to be used later to conduct more systematic searches in existing databases. We spoke to games industry and research experts in our network to request suggestions and resources, thus generating a set of 179 online relevant news articles published over the last decade [e.g., Peterson, n.d.], industry white papers [e.g., Markets and Markets, 2021], literature reviews on games for impact research [e.g., Steinkuehler & Squire, 2024], game development case studies [e.g., Schreier, 2017], textbooks [e.g., Gibson, 2022], curricula from leading game development programs at university [e.g., University of Southern California], and selected promotional materials from key game engine developers [e.g., Unity, 2005]. We then consulted an information and computer science research librarian to augment this initial set of materials, adding the results of 22 search queries using Google Scholar, ProQuest, Compendex and Inspect.

Based on this initial set of priming materials, we constructed a list of seed terms comprised of game design elements (e.g., achievements used as gamification elements in learning platforms) and technologies (e.g., game pad controllers which have been used for military drones, Peckham, 2016) that represented the key innovations that could be credited to the games industry as either (a) invented or created within the games industry directly or (b) significantly catalyzed or advanced within the games industry in some notable way; (c) key domains in which existing commercial entertainment video games were investigated or repurposed in new domains toward productive ends (e.g. Pokémon used as a health intervention) or (d) as “petri dishes” or environments for testing techniques or technologies from other disciplines (e.g., the generative testing of "swarm intelligence" robots using StarCraft: BroodWar, 1998).

The resulting set of 181 terms was then passed through ChatGPT to generate links and leads to specific source articles documenting how the games industry contributed to the key tech development under review (e.g. real-time rendering techniques). Thirty-one (31) seed terms could not be source documented and therefore were removed and the resulting set then shared again with games industry and research experts for final review and approval.

The final list included 150 seed terms across the 12 categories shown in Table 1. Nine categories were game-related tech innovations representing contributions from the games industry, and three categories were domains in which games wholesale contribute to human thriving in some way. See Appendix A for a complete list. Because inclusion required publicly citable documentation, some widely used but poorly documented technology/platform transfers may be underrepresented; where found, we coded them explicitly. Where platform-level impacts were suggested but lacked eligible documentation, we flagged them for future evidence-gathering rather than including them as findings.

Table 1: Seed Term Categories

Seed Term Category	# Publications
Serious Games	26,180
Input Devices	7516
Game Design Elements	6660
First Order Impacts	4392
Augmented and Virtual Reality (AR/VR)	3879
Game Engines	1493
Motion Capture	441
Physics Engines	60
Artificial Intelligence	0
Graphics & Rendering Techniques	0
Online and Networked Technologies	0
Petri Dish (for Primary Research)	0
	51,161

2.2.2 Queries

A team of 11 researchers (2 principal investigators, 9 trained research interns) then used the final list of 168 seed terms as the basis for individual queries in the Academic Search Complete, a multidisciplinary database indexing service that includes over 10,000 publications across academic journals, trade publications, government documents, book chapters, news articles, dissertations, books, and reports. From the initial 150 seed terms used, a total of 51,161 relevant publications were then identified for closer review. For some game-related technologies, such as those associated with innovations in artificial intelligence (AI) or specific graphic and rendering techniques that were catalyzed by the games industry, no publications were found using the technology as search term; however, this did not necessarily preclude their appearing in the data corpus as a result of other search terms. For example, artificial intelligence related search terms did not yield any findings themselves but publications on the topic were indeed found using other search terms such as corporate training.

2.2.3 Evaluation

The team of 11 researchers then reviewed the initial corpus of 51,161 articles, reports, essays, and other texts for relevance and main area of impact. The entire team met weekly for over six months during which analysis occurred. In the initial meetings, the two lead researchers led discussion and collective analysis of 10-12 articles collectively to model decision-making as to the relevance of the article (based on metadata, abstract, and key claims) and to calibrate each team member's overall judgement as to its main area of impact [Nadav, 2020; Savov, 2023].

Table 2: Domains of Impact

	# Publications		# Publications
First Order Impacts		*Health*	
Cognition	97	Exergaming	269
Well-Being	75	Mental Health	251
Social	75	Managing Chronic Conditions	243
Physical Health	24	Medical Education	205
Hand-Eye Coordination	20	Rehabilitation	178
Economic	8	Healthy Aging	80
Education		Neurological Disorders	67
General Skills	311	Lifestyle Changes	60
STEM	159	Diagnostics	45
Literacy	94	Healthcare Management	29
Health Literacy	88	Pain Management	14
Higher Education	81	Architecture, Engineering & Construction	168
Mathematics	79	Military & Defense	149
Computer Science	51	Cultural Heritage & Tourism	118
Special Education	51	Marketing & Advertising	107
Assessment	49	Urban Planning	83
Second Language Learning	49	Scientific Research & Data Visualization	79
Teacher Education	41	Environment & Sustainability	66
History & Social Studies	38	Automotive Industry	43
Brain Training	25	Disaster Preparation/Response	36
Physical Education	16	Art & Other Entertainment Domains	24
Art & Music	13	Artificial Intelligence (AI)	22
Social-Emotional Learning	4	Cybersecurity	21
Social Impact	213	Graphics & Rendering Techniques	2
Corporate Training	208		

Once assessment of the articles converged toward a shared interpretation across team members (after roughly four weeks), the corpus of 51,161 articles was divided across individual team members with at least two researchers independently deciding (a) whether the text was indeed relevant to the study and, if so, (b) to what domain or field(s) the game-related tech contributed. Additional leads, where applicable, were also noted and submitted for subsequent team evaluation for potential inclusion. Reports detailing contributions to more than one domain were coded as such. Of the entire corpus examined, a final pool of 3,579 references was determined to be relevant and substantive and then added to *Zotero*, an open-source reference management software, and organized into subfolders representing the specified domains of impact (Table 2). To avoid over-fragmentation, closely related subtopics were grouped under a small set of named categories such as *education*, *informal learning*, *gamification* (operationalized per widely used definitions in HCI/IS literature; excludes gambling and esports as distinct domains), and *technology/platform* that were defined a priori following conventions in the literature (e.g., Deterding et al., 2011; Hamari et al., 2014) and used consistently across coding and visualization. Lead researchers resolved any disputes in inclusion decisions and domain coding at weekly team meetings where team members compared observations and discussed emergent findings.

The two lead researchers then reviewed each domain subfolder within Zotero using the constant comparative method [Glaser & Strauss, 1967] to ensure internal coherence and consistency within each domain of impact the folder represented, to name the domains in ways reflecting common usage in the professions (e.g. Architecture, Engineering & Construction as a single domain), and to create operational definitions for each. Table 2 presents the final list of domains that game-related technologies have impacted and the number of publications supporting each. The full database of articles, organized by domain of contribution, can be found at https://www.zotero.org/groups/5050768/the_spread_of_ game_innovations.

Researchers tagged each report for any specific contributions to society and noted findings with respect to the affordances of games. Examples in which game technologies made clear, significant contributions beyond standard practice were tagged for potential to inform our understanding of the core affordances of game technologies. Researchers sought examples in which game technologies "request, demand, allow, encourage, discourage, and refuse" use and evidence for how users' perception of game technologies, dexterity with them, and the evolving cultural and institutional legitimacy of games more broadly shaped activity. In what follows, we report absolute counts alongside visuals, and where a category appears visually small but numerically substantial, we call this out in text to prevent misinterpretation. For example, *social impact* includes 213 publications in our corpus (see Table 2), exceeding several education sub-areas (e.g., computer science, math, and even STEM as a composite).

2.2.4 Challenges and Limitations

Throughout this project, the research team operated by a logic of achieving consensus around the primary contribution of the report and the research process was designed eliminate "false positives." Across each phase, from initial generation to final inclusion and coding, multiple researchers examined each report, resulting in at last four individuals assessing the merit of the document for inclusion in the corpus. This process was used to ensure that the final estimate of the impact of games innovations to other domains would be a conservative one. However, such an approach leaves open the risk that key contributions might also be inadvertently overlooked and under-represented. Several challenges contribute.

First, searching a domain space with undetermined boundaries can lead to utter blind spots because researchers may not be aware of a seed term (game tech innovation) or domain of activity to look for. The same problem may also result in deflated estimates of the scale of contribution. Second, not all uses of games for purposes other than entertainment are documented in published reports because not all uses of games are considered newsworthy or publishable. In this way, something as commonplace as, for example, the regular use of ST Math [1998], a game-based mathematics learning platform, in elementary classrooms, could easily go unnoted. Conversely, defense applications of game technologies are widespread but also often secretive, proprietary or otherwise largely unpublicized [Mead, 2013]. Third and finally, the frequency of reports in the literature of a given game innovation, company, or design having some purported impact in another field is not necessarily a reliable measure of that impact. Moreover, the reports themselves may not be themselves may not necessarily be evidence or reliable [Mayer, 2019]. Such risks are unfortunately endemic to a project of this scale and ambition, but since no such broad review has even been done, we judged the risks to outweigh the contribution of such a study and chose to err on the side of under-estimations.

More broadly, as with all systematic reviews, this study's findings reflect the boundaries of the search strategy and inclusion criteria. Although the methodology encompassed games as both products/systems and technologies/platforms, the resulting corpus primarily reflects academic engagement with games in applied, educational, and social contexts. Broader technological spillovers—such as the influence of game technologies on artificial intelligence, motion-picture visual effects, and other computing industries—were included only when directly documented in eligible sources. Our approach required that all cases be supported by published, citable sources—a conservative decision that favored verifiable evidence over speculative accounts. As a result, some forms of applied gaming that occur outside academic or trade publication channels may be underrepresented in the dataset—not because they do not exist or have had limited impact but simply due to the lack or published papers and reports.

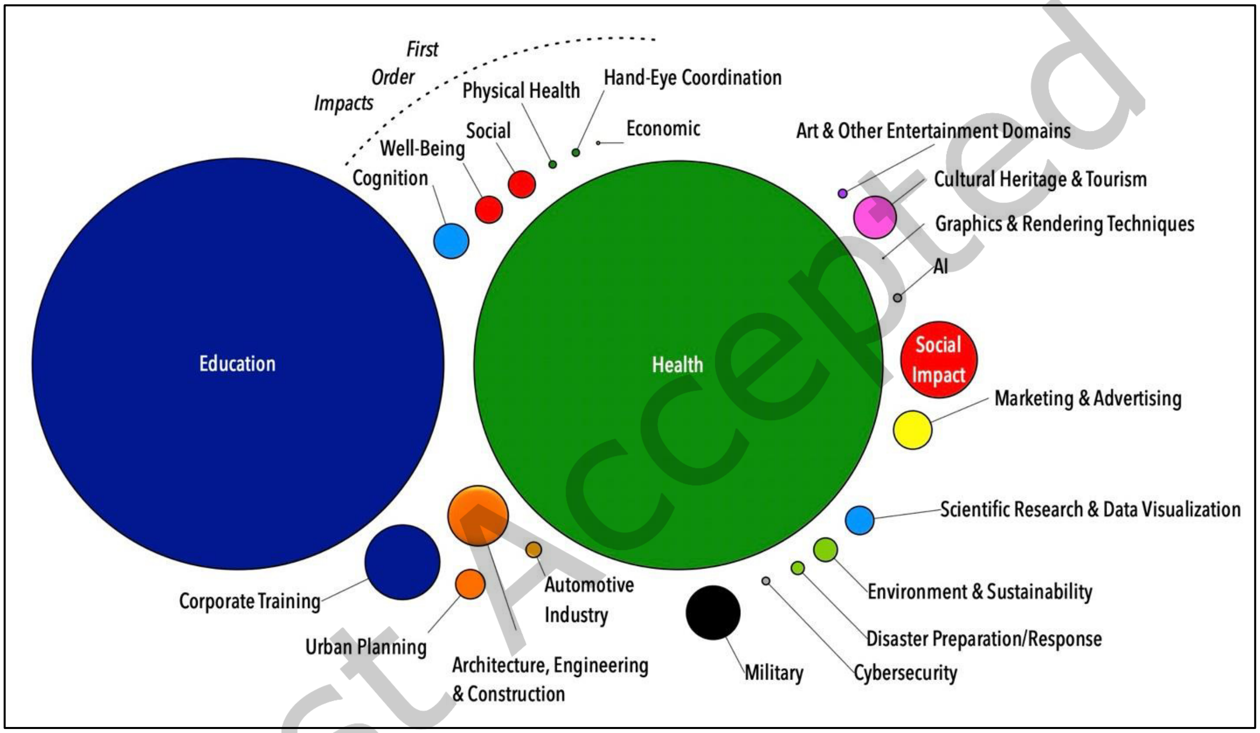

Figure 2: Bubble Chart of the Relative Contributions of Game-Based Innovations to Other Domains

3. SUMMARY OF FINDINGS

3.1. Domains of Impact

Figure 2 depicts the relative contributions of game-based innovations to the domains listed in above in Table 3. As both figure and table highlight, the majority of reported contributions were in human development, specifically education and health. It is worth noting that both sectors have long standing game-related conferences such as the Games+Learning+Society Conference (since 2005) and Games for Health (since 2004), which both reflects and potentially shapes the high number of reports found in just these two domains. And while it is true that Academic Search Complete, the database used for this study, includes publications across both academic and industry that together represents a wide span of disciplines with periodicals ranging from Architectural Digest to Military Review, a word of caution is again in order in interpreting the overall numbers and proportional representation. After all, one of the oldest and largest ostensibly "games-for-impact" conferences is the Interservice/Industry Training, Simulation and Education Conference (I/ITSEC), an event hosted annually by the miliary industrial complex, a longtime sponsor of games writ large (Mead, 2013). Below we review and summarize selected domains and subdomains of findings.

3.2 First Order Impacts

We begin our discussion with first order impacts of games or reports of direct benefits of playing off-the-shelf commercial games on cognition, well-being, social life, physical health, hand-eye coordination, and the economy (Figure 3). Such benefits are rendered directly as a result of gameplay rather than via technological innovations borrowed or transferred beyond the games industry itself.

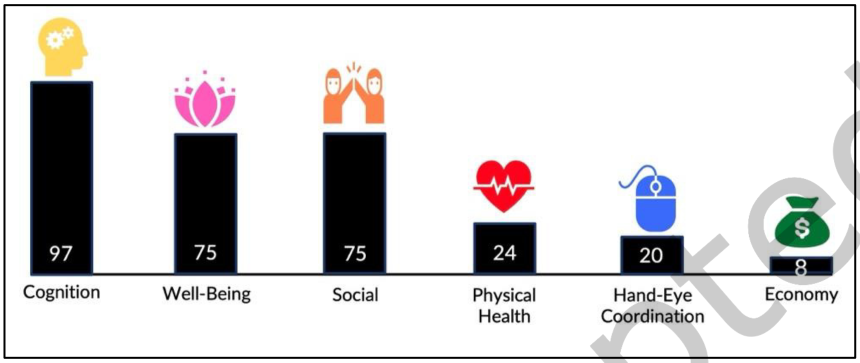

Figure 3: Number of Reports of First Order Impact of Game-Based Innovations by Domain

3.2.1 Cognition (97)

Some of the oldest and most well-established research studies on the impact of games documents the positive effects of action games on cognition, specifically attention, perception, working memory, processing speed, task switching flexibility, and the visual rotation of objects [Bavelier & Green, 2016]. People trained on action video games outperform those playing puzzle games or in control conditions [Bediou et al., 2018]. Action games require players to "learn to learn," which may be the operational mechanism [Zhang et al., 2021]. Further evidence for games efficacy exists in diagnosing, monitoring, and improving cognition for the elderly (Sirály et al., 2015), for those recovering from traumatic brain injury [Vakili et al., 2016] and in treating schizophrenia [Quiles & Verdoux, 2023]. Reflective of general trends in psychology, research on games and cognition extends to well-being, mental health, and health care because the benefit of complex cognitive activity is also connected to healthy aging, well-being, and the prevention of diseases such as Alzheimer's [Yen & Chiu, 2021]. Many such studies involve exergames or games such as Pokémon Go [2016].

3.2.2 Well-Being (75)

Well-being has cognitive, social, and emotional dimensions and is associated with both an absence of negative qualities (such as freedom from disease) and the presence of positive qualities such as social connection, positive outlook, and physical or mental fitness [Halbrook et al., 2019]. Although excessive game play can be harmful, moderate game play, particularly in the context of music role-playing and survival horror games, is associated with positive well-being [Hazel et al., 2022]. The positive impacts of Pokémon Go [2016] and similar locative AR games is a particularly well-studied phenomenon, often highlighting its benefits when played in healthy social contexts [Colder Carras et al., 2018]. Indeed, across multiple games titles and genres, the social context of games (how and with whom they are played) mediates their impacts, with positive social factors such as participation in in-game social groups such as clans or guilds mediating wellness-related outcomes even in the unlikeliest of contexts such as first-person shooters [Reer & Kraimer, 2018].

3.2.3 Social (75)

Although video games are often stereotyped as anti-social, empirical studies have demonstrated that gamers frequently play together rather than alone, with gameplay driving both competitive and collaborative behaviors [Herodotou et al., 2014]. Specific genres, such as massively multiplayer games [Zhang & Kaufman, 2017] or locative AR games (like Pokémon Go [Vaterlaus et al., 2019]) are particularly associated with pro-social outcomes. Group gaming can strengthen communication skills and resourcefulness [Hulick, 2017]. Playing team-based games can extend bounded generalized reciprocity, meaning, successful team-based game experiences can increase players' generosity even toward people not on their team [Velez, 2015]. Some of the most robust findings emanate from senior communities where the playfulness of games is associated with positive social outcomes and healthy aging [Nimrod, 2011].

3.2.4 Physical Health (24)

Playing games that require physical exertion can improve physical health. Experienced players of commercial off-the-shelf physical action games such as Dance Dance Revolution (DDR) exhibit better metabolic fitness than casual game players across a variety of measures [Sell et al., 2008]. Studies of Pokémon Go [2016] dominate research on the positive impacts of commercial gaming on physical health [Ni et al., 2019], with intentional health motivations behind gameplay amplifying its impact, a so-called "Pikachu effect" [Kaczmarek et al., 2017]. During the recent COVID pandemic, such games offered positive exercise opportunities during time of mandatory indoor isolation [Santos et al., 2021]. Bespoke exergames and exergame programs are discussed in greater depth below.

3.2.5 Hand-eye coordination (20)

Video games such as Super Monkey Ball [Nagoshi, 2001] and titles on the Nintendo Wii that recruit complex hand-eye coordinated movements [Lohse et al., 2016], which have been shown to significantly improve the performance in surgery or other medical procedures [Giannotti, 2013; Rosser Gentile et al., 2012]. Similar games have also proven effective in treating balance and agility issues among children with developmental coordination disorders [Jelsma, 2023] and even treating lazy eye among developing children [Parrish, 2005].

3.2.6 Economic (8)

As a $56 billion U.S. industry that can claim two-thirds of the country as regular players, video games clearly have economic impact [Entertainment Software Association, 2023]. The eight returns listed here pertain to the economic impact of games beyond simply listing sales figures. They include descriptions of the economic impact of games on other sectors [Stewart & Miscuraca, 2023], analyses of the human capital created by the games industry [Earp et al., 2014], and how the traditional sports industry (events and broadcasts) are leveraging digital game technologies to impact participation in sports fandom [Bowman, 2023].

3.2.7 Summary

Within a single generation, attitudes toward video games have evolved from negative stereotypes (as violent, anti-social., and addictive) to consideration of the medium equally as more or less benign, if not potentially positive, as evidence by myriad reports of their direct benefits (first order impacts) across multiple sectors. A common theme across subcategories of research on the direct impacts of games is an emphasis on play and its ability to create novel liminal experiences for players. Examining such findings, one might anticipate less emphasis on particular game titles and more emphasis on the qualities of the game experience that constitutes genuine play. To date, however, the majority of impact thus documented remain attributed to single cases (games) rather than broader genres of games, with the exception of action games and their generalized positive impact on cognition and "learning to learn."

3.3. Second Order & Tertiary Impacts

3.3.1 Education

The largest broad area of impact for games and game-based innovations is, by all measures, education. With 742 reports of impact across 16 subcategories (Figure 4), educational games and games modified or repurposed for learning in both formal and informal settings dominate discussions of the positive ends to which game-based technologies are put to use. This should not surprise us given the deep resonance between designing games and designing pedagogy, with both aimed at engaging the player or student in complex systems in ways that render the system something to think with, not only about [Gee, 2004]. Reports on the design and use of games for K-12 exceeds the use of games in higher education by an order of magnitude, with a significant body of work dedicated to understanding how specific game design elements and strategies can improve the learning process (see General Skills).

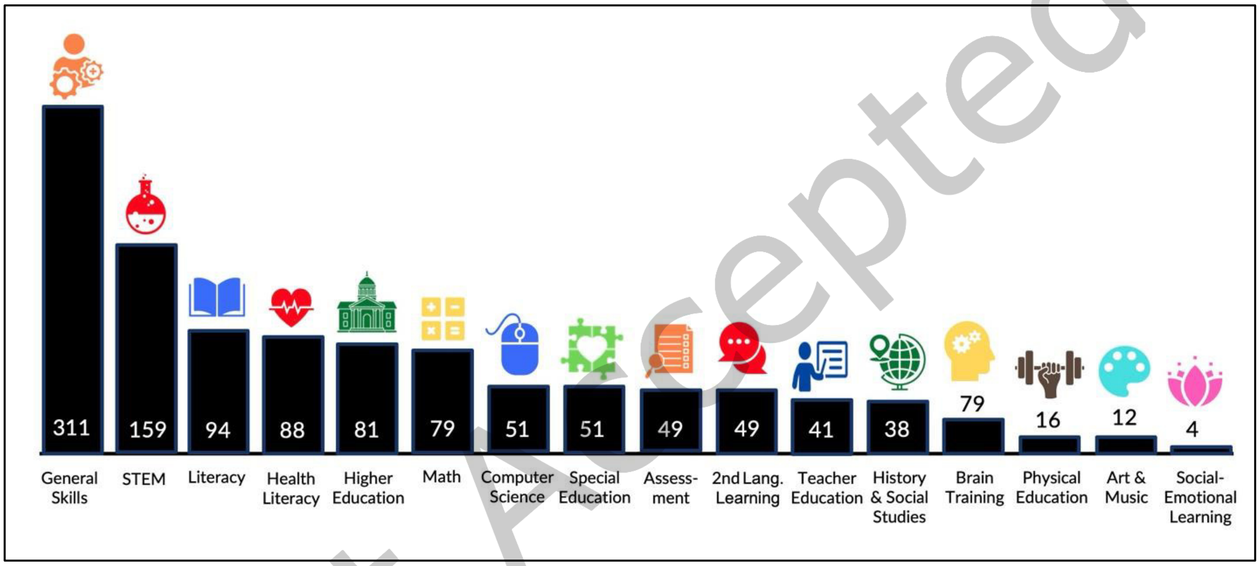

Figure 4: Number of Reported Impacts of Game-Based Innovations in Education by Subdomain

3.3.1.1 General Skills (311)

In contrast to reports on the first order impacts of games in areas like cognition, hand-eye coordination, and sociability, reports in education include significant work focusing on the contributions of specific game design elements and strategies to the learning process rather than the impacts of single game titles. For example, a single study investigated how educational gaming, social networking, gamification, and social gamification approaches compared in terms of improving student performance [de-Marcos et al., 2016], finding that, while all approaches significantly improved learning, social gamification showed greater returns. Common outcomes across reports within this subdomain include improved engagement [Breien & Wasson, 2021], motivation [Waweru et al., 2021], generalized academic achievement (such as increased grades [Kaya, & Ercag, 2023]) and behavioral indicators of compliance or such as attendance [Caton & Greenhill, 2021]. Gamification was found to be one of the more common design patterns studied and found broadly effective [Kim & Castelli, 2021]; other key game design elements investigated include interactive narrative [Bai et al., 2022], avatars [Shonfeld & Greenstein, 2021], and roleplay [Poy et al., 2019]. Instructional designers have often asked, “What are games particularly good for?” and in this review, evidence was found for games’ positive impact on a wide range of educational., outcomes across diverse contexts, with effective designers commonly combining and synthesizing multiple techniques, suggesting that questions of “whether to use games" may rely more on situational constraints, contexts, and user preferences rather than games as a broad medium.

Also within this subdomain are games targeting general skills overtly. While the majority of educational games target specific academic domains, as the organization of this broad section of educational impacts demonstrates, it is not uncommon to find games designed and used for general life skills or so-called 21st century skills [Qian & Clark, 2016]. Good game play often consists of making choices, for example, and thus games have been widely used to develop just such skills. Common targeted skills include critical thinking [Lee et al., 2016]. problem solving [Chang et al., 2017], remediation of cognitive bias [Dunbar et al., 2017], ethical reasoning [Nardo & Gaydos, 2021], and early elementary skills such as basic academic vocabular or numeracy [Ponticorvo et al., 2017]. Reports of games' impact on general skills emphasize less the technical affordances of games and more their design features and patterns.

3.3.1.2 STEM (159)

Games have been used across almost every area of science, technology, engineering and mathematics (STEM) to address learning goals including content understanding, scientific literacy, and identification with science [National Research Council, 2011]. Here we discuss games for mathematics and for computer science separately. Federal investments in science education have led to significant innovations in games for learning [Clark, et al., 2023], interest in novel methods to engage youth in science [Leonard et al., 2016], novel approaches to improving academic performance and motivation in engineering [Pando Cerra et al., 2022], and the importance of innovations to address concepts such as scientific citizenship [Gaydos & Squire, 2012]. Multiple meta-analyses now demonstrate that games can outperform other educational methods in science, particularly when scaffolded through embedded instructional techniques such as tutors or analytics [Cloude et al., 2021]. Science educators have used games to increase not only scientific knowledge [Martin et al., 2019] but also scientific argumentation [Bağ & Çalık, 2021], sense-making across multiple representations [Virk et al., 2015], facility with scientific tools, increased identification with the enterprise of science [Sun et al., 2021], and simply discovering nature with mobile games [Goodwin, 2016]. Such studies reveal that a strength of video games lies in their interactivity which enables designers to embed scaffolded learning experiences [Clark et al., 2016]. An ongoing challenge for science educators using educational games in classrooms is the publication of such games, their integration into the classroom, and their sustained support once initial project funding has ended given the economic realities of school purchasing and how and how such decisions are made [Ito, 2009].

3.3.1.3Literacy (94)

Games have been used to promote a range of literacy, here defined broadly as the ability to make and communicate meaning out of oral language and print text effectively. Literacy researchers, practitioners and developers have explored how games promote literacy engagement, the role of games in today's literacy multimodal environment, and how games might be designed to scaffold and support literacy outright. For example, students have used Minecraft [Persson, 2011] to recreate scenes from literature which led to deeper connections with fiction [Marlatt, 2018]. Teachers have used other commercial off-the-shelf games as writing prompts, enabling students to improve writing by analyzing games as texts [Stufft & von Gillern, 2021]. Bespoke games have also been developed to target literacy directly; for example, the multiplayer online game Rochester Castle designed to promote language learning in a school in Western Australia [Lee et al., 2005]. Several such games target students with learning disabilities, using core features of games (active challenges, interactivity, fun feedback) to increase engagement to enhance learning [Shohieb et al., 2022]. Increased engagement in the underlying activity (here, literacy) as a result of game play is a pattern worth examining in other domains.

3.3.1.4 Health Literacy (88)

Games are likewise used to increase health literacy in non-clinical contexts across a wide range of age, targeting topics groups that range from smoking cessation [Heffner et al., 2021] to safe sexual behaviors [Alencar et al., 2022], from oral care [Aljafari et al., 2022] to infant care [D'Agostini, 2020], from road safety [Khan, 2021] to motor movements [Hammond et al., 2014], from general health literacy [Parisod et al., 2017] to vaccine use [Montagni et al., 2020]. A published review of 1,452 game-based health programs [Primack et al., 2012] found 38 titles with demonstrated impact across 195 health outcomes, concluding that "video games improved 69% of psychological therapy outcomes, 59% of physical therapy outcomes, 50% of physical activity outcomes, 46% of clinician skills outcomes, 42% of health education outcomes, 42% of pain distraction outcomes, and 37% of disease self-management outcomes" (p.630). In sum, games for health literacy demonstrate clear changes in health-related outcomes.

3.3.1.5 Higher Education (81)

Games in higher education often focus on professional education in area such as business, information management, video game development, manufacturing, or nursing education. Games facilitate strategic problem solving and leadership training and structure outside-of-class experiences driven by local needs rather than theoretical concerns [Khaldi et al., 2023]; gamifying orientation days is one such prototypical use [Vanderstraeten, et al., 2022]. Other common design patterns include role play in professional scenarios and cases [Rixon & Skipton, 2018], motivational structures to influence out-of-(college)-classroom learning experiences [Torrado Cespon, & Díaz Lage, 2022], and design features such as avatars to improve eLearning environments [Hu et al., 2023]. Higher education games often seek to restructure the grammar of the classroom by gamifying structures (e.g., student as role player, activities as quests) rather than gamifying content (e.g., playing a game to learn virology)[Kahldi et al., 2023]. When the overarching activity structure is changed to that of a game, dramatic changes can occur.

3.3.1.6 Mathematics (79)

Game-based pedagogies are well established in mathematics education. Math Blaster [Davidson & Eckert, 1983] is a canonical title in math education, selling over one million copies and launching a company that in time became the parent company for Blizzard Entertainment [Ito, 2009]. Like Math Blaster, most math education games have focused on operations (including fractions, [Masek et al., 2017]), designed more to facilitate practice and rarely grounded in learning theory [Tokac et al., 2019]. A more recent generation of game-based learning products such as ST Math [MIND Research Institute, 1998] began as research-based interventions and now have years of research and development behind them as well as demonstrated impact across multiple schools and districts [Rutherford et al., 2009]. Meta-analyses that investigate the learning outcomes find that games can outperform other instructional approaches, although counterexamples exist in which well-designed traditional techniques outperform games (Ersen & Ergül, 2022; Tokac et al., 2019]. In their discussion of the research on such programs, Carolan and Zielezinski [2019] advocate for efficacy portfolios in understanding impact, ones that includes a defining logic model, formative research, and summative research. Successful math education projects represent phenomenon across time and space [Bodner & Coulson, 2021], improve working memory [Ramani et al., 2020], serve diverse learners [Cooper, 2018], and address affective issues and not just conceptual ones [Huang et al., 2014]. However, the particulars of how a game is designed, implemented, and evaluated matter more than it simply "being a game."

3.3.1.7 Computer Science (51)

Games are also used as media for teaching, tools for teaching specific concepts, assessments, and a structure for organizing curricula in computer science. Constructionist environments such as Scratch [2003], Kodu [2009], or Minecraft [Persson, 2011] are used for teaching programming, with projects such as MAGDAIRE based on such environments (here, Kodu) demonstrating clear learning gains on post measures [Uluay & Dogan, 2020]. Noting the limitations of pre- and post-tests for measuring learning in programming, game-based learning environments have been created that seek to introduce computer science constructs in ways that enable practice in complex situations and then assess learner performance [Weintrop & Wilensky, 2014]. Games for teaching computer science constructs, such as the puzzle game while True: learn(), (2019), are entertainment products in the consumer marketplace [Lindberg et al., 2019]. Finally, using games as a method for organizing activities through gamification or badges can improve motivation and engagement for learning programming language concepts [Abu-Hammad & Hamtini, 2023].

3.3.1.8 Assessment (49)

Games have emerged as an alternative to tests for assessment [Delacruz et al., 2010]. Games are both information-rich problem-solving environments in which learning can be assessed and models for documenting achievement through systems such as badging. CyGaMES [Reese, 2015], for example, is a lunar construction game with a robust assessment backend that enables embedded assessments, learning analytics, game data reporting and even real-time online visualizations of play data. Newton's Playground [Shute et a., 2013] also tracks learners' behaviors to enable "stealth" assessments, which can award badges as an alternative credentialling structure [McDaniel & Fanfarelli, 2016]. Games offer new opportunities for at-risk students and students with disabilities, and they have been validated for use in situations in which the constraints of testing scenarios (e.g., time constraints) threaten the validity outcomes of traditional tests [Marino et al., 2011]. Evidence even suggests that games may be better assessments than pen-and-paper tests; a case in point is Shadowspect, a reliable and validated test for spatial skills [Kim, et al., 2023].

3.3.1.9 Other Contribution to Education (237)

In addition to these core subdomains in Education, games are also used in Special Education (51 reports), Second Language Learning (49), Teacher Education (41), History & Social Studies (38), Brain Training (25), Physical Education (16), Art & Music (13) Social-Emotional Learning (4). Reports range from dance games in physical education classes [Gibbs et al., 2017] to students modding [Andersson, 2013] in social studies classrooms [Loban 2021]. From these myriad domains emerge additional themes, including: games as tools for supporting fundamental skill and knowledge development [Wang, 2023], games as constructionist contexts for students (and teachers) to create with [Esteve-González et al., 2016], and gamification systems that structure the curriculum [Pingmuang & Koraneekij, 2022]. The impact of games can be found across all ages and areas of education, in varying degrees and to varying extents, so that no area is entirely untouched by the medium.

3.3.2 Health

Games have been used for a range of purposes in health care and medical training, as well documented in games for health conferences, journals and initiatives [Wattanasoontorn et al., 2013]. They have been proposed as a unifying structure for changing lifestyle behaviors (diet, exercise) while tracking physiological data so that patients can take control over their health care, communicate better with health care providers, gain assistance in rehabilitation, and ultimately, have improved healthcare delivery [McCallum, 2012]. Games are also used in specifically in health diagnostics, the management of chronic health conditions and pain, training for surgery and other medical procedures, and even the management of healthcare services. Although historically associated with sedentary lifestyles and poor physical health [Puolitaival et al., 2020], video games promoting physical movement (e.g., Pokemon Go [2016]) through motion-based controllers (e.g., Nintendo Wii [2006]) and games' tracking techniques (e.g., Zombies, Run! [2012]) open new opportunities for games to improve physical health. The impact of such innovations on players have been explored, documented, and reported in the news [Silverman, 2022], case studies [Rhu et al., 2022], and meta-analyses [Peng et al., 2011]. Games can improve physical health among children [Cai et al., 2017], older adults

[Rice, 2022], and the general public [Berkovsky et al., 2012] for healthy individuals, populations vulnerable to disease, and those facing chronic or acute conditions.

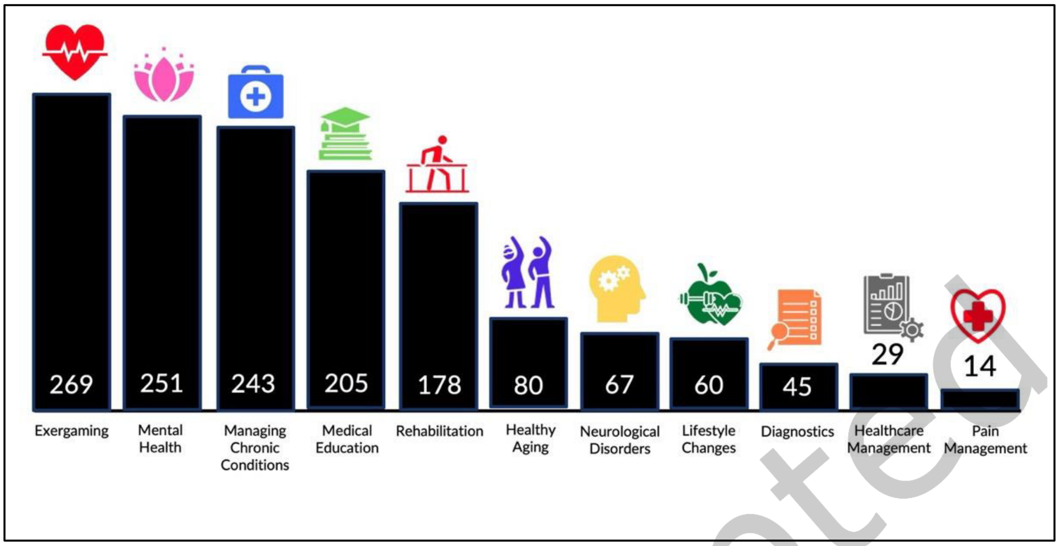

Figure 5: Number of Reported Impacts of Game-Based Innovations in Health by Subdomain

3.3.2.1 Exergaming (269)

Unlike education or medicine, where games typically often originate in academics and then enter the mainstream, exergames often begin in the entertainment marketplace. Games such as Dance, Dance Revolution [1998], Wii Sports [Morii, 2006], and Pokemon Go [2016] with mainstream popularity have inspired physical education professionals to consider their potential impact, which in turn has led to vibrant domain of programmatic activities and research. Exergames are used in homes, schools, after-schools, hospitals, and elderly care contexts for increasing physical activity to improve metabolic functioning, balance, postural stability, agility, or mental health [Bleakley, et al., 2015; Liu et al., 2020]. Research on exergaming [e.g., Ma, et al., 2018] reveals that exergames provide light to moderate intensity activity for users, able to increase physical activity, improve motivation, and strengthen compliance with exercise regiments; still, they are not a panacea: user choice, motivation, and contextual factors all play a role in shaping their adoption [Peng et al., 2013] and activity levels often return to baseline once the intervention has ended [Li et al., 2021]. Regardless, the breadth of activity in this sector is striking, including not just single studies but broad literature reviews and meta-analyses evidencing robust areas of research.

3.3.2.2 Mental Health (251)

Reports of games to improve mental health predominantly include (1) examinations of how people use games to manage mental health, including intrusive memories of traumatic events, depression, or anxiety, and (2) games specifically designed to calm the nervous system or improve mental health in otherwise healthy functioning individuals. Games may be associated with negative mental states in the popular imagination, but for many, games are a respite from the world and provide comfort, soothing, and relaxation [Project Horseshoe, 2017]. During the pandemic, for example, many players turned to Animal Crossing [2001] for its cute, cozy aesthetics, sense of order, and ability to connect to other gamers remotely [Lewis et al., 2021]. Generations of players have turned to Tetris [Pajitnov, 1985] to reduce symptoms of stress, and current neuroscientific investigations bearing such anecdotal reports out: Butler, et al. [2020] found that Tetris gameplay, compared to controls, increased hippocampal volume which was, in turn, correlated with reductions in symptoms of PTSD, depression and anxiety. Cute characters and pleasing colors can calm the nervous system [Rupp et al., 2017] and designers have begun integrating just such features into apps specifically designed to calm patients [Astle & Writer, 2022]. Participants with social anxieties or autism report that avatar-mediated interactions can facilitate sharing and further promote healthy effects [Yoshida et al., 2022]. Although turning to games as a form of emotional or mental relief runs risk of addiction, the evidence on positive gaming suggests that it can also be a positive form of respite.

3.3.2.3 Managing Chronic Conditions (243)

Games are also used in chronic disease management such cystic fibrosis, diabetes, and multiple sclerosis. As the links between behaviors, health, and lifestyle are better understood, games offer attractive options for establishing habits that help people manage their health conditions; a meta-review revealed that games were effective in improving knowledge, behavior, and clinical outcomes [Charlier et al., 2016]. The pharmaceutical company Pfizer partnered with Drexel University to create Hemocraft, a spin on the ultra-successful Minecraft [Persson, 2011] in which hemophilia patients manage their healthcare via gameplay coupled with their HemMobile Striiv wearable device [Helfand, 2017]. Games to help youth manage chronic diseases have a deep history in games for change; in 1992 Raya Systems published Captain Novolin [Rodgers, 1992] for the Super Nintendo, a game whose main hero has type 1 diabetes, which led to improvements in player health and self-efficacy, increased health literacy motivations, and improvements in communication [Lieberman, 1997]. Other titles in the series include Rex Ronan [Condor, 1994] for tobacco cessation, Packy and Marlon [1995] for Diabetes mellitus type 1, and Bronckie the Bronchiasaurus [Manriquez, 1995] for asthma, all of which have demonstrated positive impacts among players [Lieberman, 2001]. A review of serious games in asthma education found increases in knowledge but less evidence for health outcomes across studies, [Drummond et al., 2017]. With diseases such as cystic fibrosis, games help patients understand the reality of their health conditions and experience agency in fighting disease [Vagg et al., 2019]. When people with chronic conditions play games that increase their physical activity levels, indicators such as BMI, hemoglobin A1c levels, and obesity levels are improved [Bossen et al. 2020; Patel, et al., 2021]. Leverage points for such games appears to be their ability to address knowledge, attitudes, and behavioral change in an integrated format; their relative cost savings in reducing the need for other treatments or emergency room visits, and player time commitment. Generally, however, investment in game-based approaches makes sense in situations with lifelong health concerns.

3.3.2.4 Medical Education (205)

Games are also commonly used among health care professionals to improve their own knowledge, skills, or awareness in medical and health classrooms [Berger et al., 2018], medical residencies [Gue et al., 2022], and professional practice situations [Bethea et al., 2014]. The wide scope of games in health care education is evidenced by a study of simulations and games in nursing education, which found 118 studies with 6738 participants, including 55 studies on simulated patients, 40 studies on role plays, 12 on VR, 9 on manikins, and 9 on voice simulations [Piot et al. 2022], with the predominance of studies concluding that simulation-based approaches out-performed traditional ones (although the possibility for file drawer effects should be considered). Simulation-based games fit naturally in medical education, which regularly adopts new technologies, especially ones in which key procedures can be simulated.

Games are used by in fields with medical responsibilities, such as first responders, childcare providers, or nursing assistants, for professional development of new skills such as learning indicators of domestic violence [Mason & Turner, 2018] or to refresh participants' knowledge of existing procedures and deepen long-term retention such as newborn resuscitation skills [Bardelli, et al., 2022]. A particularly compelling use of games is for emergency field procedures, such as soldiers practicing field medicine on animals during combat ("ECS adds canine interventions," 2023) or games to help children learn to resuscitate other children [Semeraro, et al., 2017].

Games also enhance medical procedures such surgeries [Chalhoub et al., 2018], blood transfusions [Tan et al., 2017], and ultrasounds [McGaffey et al., 2023]. A decade of research has identified a positive association between warming up for surgery with video games and subsequent performance on surgical procedures [Katria, 2022; Rosser, et al., 2012]. This body of evidence supporting the effectiveness and appeal of game-based approaches to medical procedures suggests a fit between games and complex procedures general [Lee, et al. 2018; Liu & Curet, 2015]. Games' capacity to represent complex, dynamic systems, to support learning through failure, and to compress time allows students to experience feedback and observe the consequences of decisions in low-risk situations. Cutting-edge applications increasingly integrate AI into the virtual training environment [Bourdillon et al., 2023], suggesting one particularly fruitful future domain for others to track with respect to AI-enabled game-based approaches.

3.3.2.5 Rehabilitation (178)

The confluence of new controllers (such as the *Wii Remote* [2006], *Wii Balance Board* [2007] and *Kinect* [2010]), gamification of design patterns, data tracking, and new design tools make gamified approaches to rehabilitation, a domain that requires extensive individual repetitive and routinized practice, an excellent fit across multiple subdomains. Examples include gamified approaches to rehabilitation of injuries such as metacarpal fractures [Then et al., 2020] and the use of game controllers and interfaces for rehabilitation exercises after sport-related concussions [Merchant-Born et al., 2017] and even burn injuries [Pham et al., 2018]. Thousands of research studies on games for rehabilitation have been conducted over the past two decades. Bonnechere and colleagues [2016] conducted a meta-analysis with 126 studies across over 4000 articles and found that, overall, game-based approaches performed as well as conventional therapies. A subsequent review found evidence of games significantly positively contributing to rehabilitation from stroke, cerebral palsy, and diagnosed neurological conditions [Akram et al., 2022]. Games can replace more expensive techniques, augment human interaction, recruit routine use, and tracking compliance data, all indicators of how and under what conditions games may be more broadly useful.

The largest number of reports of games for rehabilitation is for strokes (64 reports total). For example, a computer game outfitted with a custom controller to improve postural balance in stroke patients out-performed traditional techniques [Hsieh, 2019]. Similarly, *Mindpod*, a spin-off from John Hopkins Professor John Krakauer's BLAM lab, (www.blam-lab-jhu.org/john-krakauer) helps people recover from strokes by controlling a simulated dolphin and engaging carefully tasks that accelerate rewiring of the brain around damaged cells to help patients regain control of their nervous systems [Moore, 2021, ¶3]. Games appear to be particularly effective for stroke rehabilitation due to their challenge-driven nature, capacity to help users recover from errors, rich contextuality, and customizability [Mubin, et al., 2022].

The second largest number of reports of games for rehabilitation is for helping patients with cancer (36 reports) manage symptoms from chemotherapy [Lorezel, et al., 2020], received improve diagnostic imaging [Ohman et al., 2014], more effectively recover through exergaming interventions [Tough et al., 2018], self-advocate in advanced stages of the disease [Thomas et al., 2019], improve their own psychological health through recreational play [Comello et al., 2016], The game *Re:Mission* [2006] developed in 2006 by HopeLab (and its sequel including *Re-Mission 2: Nanobot's Revenge* [2013]) reveal the ability of games to increase adherence to rehabilitation protocols and bolster the psychological outlook of patients for whom psychological outlook is a crucial part of recovery.

3.3.2.6 Other Contribution to Health (295)

In addition to these core subdomains in Health, games are also used in Healthy Aging (80), Neurological Disorders (67), Lifestyle Changes (60), Diagnostics (45), Healthcare Management (29) and Pain Management (14). Reports range from games to improve crucial health behaviors from geriatric decision making related to healthcare [Lagro et al., 2014] to sustained attention and executive functions among individuals with multiple sclerosis [De Giglio et al., 2015], from gamification to increase physical among patients with elevated risk for cardiovascular disease [Fanaroff et al., 2023] to the use of NVidia graphic cards, catalyzed by the games industry, for 3D real-time parallel volume rendering for medical imaging [Asave et al., 2009], from games to teach optimal design of a medical practice (Hannig, 2012) to mobile health games for pain management among children in day surgery care [Rantala et al., 2022]. The impacts of games for health span all ages and increasingly, many areas of health care and management.

3.3.3 Social Impact (213)

Games for social impact are now commonplace [e.g. the Games for Change Festival]. Example games within this domain include When Rivers Were Trails [LaPensee, 2019]; a game about the impact of 1890s US colonization on the indigenous, Papers, Please [Pope, 2013], a game where players are inspectors at a border crossing, designed to educate people about the complex issues in migration; This War of Mine [2014], a war game in which players are civilizations (rather than soldiers) trying to escape the perils of war; and Change: A Homeless Survival Experience [Hayes & Odell, 2020] in which players must learn how to survive being homeless. With dozens of such games now on the market, issues facing this sector include how to sustainably develop and promote such titles, how to help them reach target audiences whose views the hope to change, and how to judge their impact. Such models are emerging, often in conjunction with streaming revenue; for example, for the social impact game Stray ["How the video game 'Stray' is helping," 2023], where players are a stray cat seeking safety in an alien world, publishers decided to partner with streamers to publicize the game while raising $10,000s for animal shelters. As a large, thriving sector with relatively little funding, games for social issues may be continue being an innovator in terms of both content and new business models.

3.3.4 Corporate Training (208)

While closely related to learning, education, and cognition, games for corporate job training in private contexts is its own, distinct domain with decades long history. Nearly fifty years has passed since Thiagarajan and Stolovitch's (1978) Instructional Simulation Games and games for corporate training are now mainstream, routinely used for "hard skills" or procedural knowledge [Jantakun et al., 2023], "soft skills" or interpersonal skill training [Meyer, 2014] and specific knowledge sub-domains. Games have made significant impact in compliance training, as academics who have done role playing versions of institutional review board training can attest. The University of Wisconsin-Madison's Fair Play game has been used for racial and gender bias training [Gutierrez et al., 2014]. VR technologies, which offer immersive training solutions more engaging than traditional eLearning but less extensive than face-to-face training, may further accelerate the integration of game technologies into routine corporate training practice [Bailenson, 2020]. Escape rooms [Arpin, 2022], mixed reality for teacher professional development training [Garrett et al., 2020], and augmented reality for manufacturing training ["How AR and VR are transforming training," 2018] are but three examples of emerging game-based training approaches in the workplace. Games for training succeed when they meet institutional goals, such as providing training more cheaply, asynchronously, with greater compliance, or with increased satisfaction.

3.3.5 Architecture, Engineering, & Construction (168)

Games tools are used to prototype ideas, facilitate communication across stakeholders, and visualize solutions across multiple domains including architecture (43 reports), engineering (57), construction (28), manufacturing (9), and the use of digital twins (31). Teasing out game specific innovations in companies such as AutoCad, which develops tools used in games, film, and construction, can be difficult, but the recent expansion of the use of game engines to render environments both real (digital twins) and imagined is generating significant innovations [Sydora et al., 2021; Weatherbed, 2022]. Digital twins, digital models of physical, real-world objects, systems, or processes are a compelling case in point. For example, a digital twin of the Vancouver airport gathers data about passenger demands, counts, and forecasts, and enable airport staff to anticipate issues caused by maintenance tasks and then model and forecast the impact of delays [Redins, 2022]. Game engines excel at rendering complex systems in real time, making them excellent tools for modeling interacting systems, particularly those with agents that can be depicted according to rules (such as passengers), and the consequences of decisions examined by various stakeholders, including passengers themselves [Grübel et al., 2022]. Game-like tools that can integrate real-time data have been used to improve monitoring in construction projects [Pour Rahimian et al., 2020] and game design techniques have been leveraged to collaboratively designed a nuclear power plant within a multi-disciplinary design team [Boy et al., 2016]. Game tools may be particularly good for representing rule-based systems sitting at the intersection of multiple domains requiring consumer-facing solutions rendered in real time, with digital twins serving as a model for other systems.

3.3.6 Military & Defense (149)

Given the well-documented relationship between the military and games [Mead, 2013], including high profile collaborations such America's Army [2002], Full Spectrum Warrior [McWilliams, 2004] and Rainbox Six: Rogue Spear [Schnurr, 1999], one might anticipate military and defense uses of games to far exceed others. Indeed, the annual I/ITSEC conference, which drew 20,000 participants at its peak in the early 2000s, features dozens of game-based learning, training, digital twin and demonstration projects each year. For example, in DARWARS Ambush! [2004] created with the Torque game engine [2009], multiple players complete training exercises such as participating in a convoy in Iraq while avoiding ambush [Chatham, 2007]. America's Army, the promotional and recruitment game funded by the United States Army and first published in 2002, now has 41versions and updates, leading to a dislocation of categories such as soldiers and citizens, wars and games, work and play, and virtual and reality [Davis et al., 2003]. Its success spawned a wave of similar game-based military training programs [Insinna, 2013]. In return, the military and defense industries function as a research and development context for games; examples in recent years include head mounted VR displays for space flight [Waisberg et al., 2022], mixed-reality platforms for training battlefield first aid [Du, et al., 2022], and haptics for firearm training [Wei et al., 2019]. Games have also been used to avoid conflicts; examples include games to train soldiers on cultural norms and differences [Button, 2009], de-escalate conflicts [Thilmany, 2005], and train soldiers for non-violent missions [Lin et al., 2011] by leveraging games' full-motion interactive 3D nature tools for animating humans. As games are used in the military for activities beyond specialized training (such as flying planes), better knowledge transfer between the military other domains would be beneficial. Technically, games fill a niche alongside other simulation platforms in that they enable rapid development and deployment of human-scale learning environments, such as crowded city streets used in cultural training simulations, and serve as useful platforms for prototyping new input and output mechanisms.

3.3.7 Cultural Heritage & Tourism (118)

"Visit the pyramids, ancient Rome, or the Hagia Sophia, virtually!" is a long-held vision of educators, historians, and cultural preservationists, and indeed game technologies are now used across a variety of related subdomains including cultural heritage (77), tourism (19), archeology (12), and museums (9). The creation of interactive, immersive virtual sites, such as the virtual Hagia Sophia [Doker & Kirkangicoglu, 2018] with game technologies to be deployed on computers, phones, or VR has becoming increasingly commonplace. With Minecraft [Persson, 2011], for example, anyone, even kids, can recreate world wonders in virtual spaces for others to experience [Garcia-Fernandez & Medeiros, 2019]. Animating 3D structures to suggest the feel of an important cultural space, such as the Beijing Opera House [Wang, 2009], is the kind of challenge now tackled by VR developers. The most sophisticated of such history-brought-to-life approaches may be the tours embedded in the Assassin's Creed series [2007]; written by historians and embedded within the commercial game, they guide the player through the ancient places and cultures depicted, highlighting historical accuracies, anomalies, or otherwise interesting details [Balela & Mundy, 2015]. Teacher's guides have been created of and for such materials, blurring the line between games for entertainment and for education [Berger & Staley, 2014].

Cultural heritage games also tackle more narrowed goals such as introducing players to traditional Chinese herbal medicine [Ee et al., 2018] or indigenous names for wildflowers [Rahaman et al., 2021]. Designers of cultural heritage, history, museum and tourism experiences frequently enable indigenous and first nation peoples to collaborate in representations of their histories, cultures, and belief systems [Anderson et al., 2010]. Moreover, First Nations use game tools and techniques themselves to preserve their culture, ensure their stories survive, and share their histories outside of colonizers' partial histories [CPI Office, 2021]. Much as cultural locations have websites to promote tourism and communicate with the public, one might imagine indigenously led immersive representations (much like digital twins, discussed above), to both communicate internally and with the public [Chanet al., 2020]. Ensuring that an equitable team of stakeholders, particularly otherwise marginalized voices, is critical to such projects; indeed, such practices would surely improve games across other domains.

3.3.8 Marketing & Advertising (107)

Harnessing player data for more effective marketing [Che et al., 2023], mining games for engagement strategies [Hayward, 2017], mining esports for new consumption patterns [Seo, 2013], leveraging avatars marketing strategies [Miao, 2022], and advertising directly through games themselves [Bulik, 2009; "Why Unreal Engine," n.d.] are key ways games and game-based innovations have been leveraged for marketing and advertising. Brands seeking to build identification with their products through reward structures leverage gamification techniques [Bitrián & Catalán, 2022]. Avatars offer opportunities to examine consumer desires, gather data on their choices, and present brands to them via digital worlds [Alves & Soares, 2013]. In a review of the literature, Miao and colleagues [2022] conclude that the behavioral and formal realism versus stylization are two axes that explain an avatar's marketing effectiveness. As more activities migrate to synthetic worlds, games become a potent vehicle for understanding how consumers make choices in myriad domains [Gonzales-Chávez &Vila-Lopez, 2021] and increasingly popular contexts for studying consumer choices and desires, brand identity and design.

3.3.9 Urban Planning (83)

Since the release of Sim City [Wright, 1989], computer games have been associated with city planning. Urban planning games have since broadened to include city builders, simulators, and physical games played in urban spaces on smart phones [Kolson, 1996; Stokes et al., 2021]. Urban planning games are used for visualization (24 reports); theorization of city planning, growth, and user experiences of urban landscapes (21); education and training (18), city planning processes (12), improving traffic safety (5), city health (2), and the generation of new smart city models (6). Urban planning games feature emergent events unfolding in real-time in response to player actions, which include strategic planning, resource management, and data-driven decision making, making urban planning games an important precursor to digital twins (Euklidiadas, 2022). The interactive 2D/3D nature of games combined with their capacity to represent unfolding processes makes them excellent tools for visualization of phenomena ranging from historical cityscapes [Schmidt et al., 2021] to new ecosystems services to support them [Yang & Delparte, 2022]. Urban planners and designers use game engines (particularly virtual reality) to investigate specific planning issues, such as lighting [Scorpio, et al., 2020], or the impact of planning decisions on street-level residential experience. Urban planners and the public use the popular game Sim City [Wright, 1989] to imagine innovative urban environments and dweller experiences in them, thus enabling public conversations about key issues such as zoning, parking, public transportation, parks, and environmental impacts [Sneed, 2013]. Educational efforts leverage this dynamic relationship between urban planning and the public sphere, creating events in which games simultaneously educate youth about urban planning careers while also training urban planners [Minnery & Searle, 2014]. Indeed, participatory design approaches leverage the accessibility of games to engage the public in urban planning processes [Gordon & Manosevitch, 2011], suggesting a dynamic future for games and urban planning.

3.3.10 Scientific Research & Data Visualization (79)

Games enable and platform scientific research on phenomena ranging from traffic modeling [Liao et al., 2022] to experiments in synthetic biology [Matzko et al., 2023], from geophysical systems [Wang, et al., 2020] to molecules [Lv et al., 2013]. The protein folding game, Fold.it [2008] is the most known and influential of such games, having contributed to scientific discoveries in the nature of both protein folding [Eiben, 2012] and collaborative problem solving [Bauer, Flatten, & Popovic, 2017]. Subsequent knowledge games, games that seek to contribute to scientific understandings as well as general entertainment [Schrier, 2016] have made significant contributions to brain mapping [Langston, 2017], "molecular modeling, sequence alignment, neuroscience, pathology, cellular biology, genomics, and human cognition" [Das et al., 2019, p.253]. Video game technologies have also been employed for scientific visualization [Ly et al., 2013], with innovative platforms such as iBET [Nakano, et al., 2016] now able to allow export of scientific models into VR systems such as the Oculus Rift via the game engine Unity [2005]. Similar processes have been used in synthetic biology and chemical engineering with the Unreal game engine [Sweeney, 1995], enabling scientists to scale up from single-cell to multi-cell simulations [Matzko et al., 2023]. Such projects highlight the accessibility of game engines and the robust nature of game communities in making important contributions to cutting-edge scientific discovery.

3.3.11 Environment & Sustainability (66)

Games in environmental science and sustainability include research and development projects to improve public understanding of environmental issues (26 reports), to understanding environmental systems and the impact of human intervention (such as studying the impact of regulations on carbon emissions) (16), to change behavior through gamification and other game-based strategies toward more sustainable practices (12), and analyses of the mechanisms by which games make an impact on the environment and our understandings of it (12). Many game projects span multiple foci; for example, the University of Washington's EarthGames [Lawler, 2015] teaches undergraduates about environmental issues with games, educates the public through games, and researches the game's impact on player behaviors [Kwok, 2019]. Environmental engineers, researchers, and managers have begun using game technologies to model ecological phenomena, such as algae blooms [Bryceson et al., 2022], ecosystems such as wetlands [Lu et al., 2023], water resources in drought zones [Wang & Davies, 2015], carbon emissions policy [Lee & Hussain, 2022], or cattle farms [Gómez-Prada & Gómez-Sandoval, 2019]. Gamified approaches have enabled users to set goals, track behaviors, and change practices to reduce their environmental impact [Nastis & Pagoni, 2019]. The dream of simultaneously modeling complex systems, testing policy decisions, improving public understanding, and researching impact is endemic of cutting-edge work in sustainability [Bell-Gawne, et al., 2013; Tomlinson et al., 2008].

3.3.12 Automotive Industries (43)

Games-based innovations have contributed to automotive design (13 reports), design of the driving experience (9), methods for managing traffic (8), cockpit design (7), and safety (6). Although companies use specialized software for designing automobiles, game-related tools support consumer-facing applications useful for marketing, user research, and engaging multiple stakeholders [Beecham, 2016]. Games tools' integration with VR platforms has further driven integration in automotives [Crosse, 2021]. They have also recently been employed to reconceptualize the cockpit experience ["Automotive smart cockpit design," 2022]; as automobiles employ more data streams and tools, designers have begun to leverage game interfaces, elements, and design techniques to enhance the driving experience [Leggett, 2022]. Opportunities also exist for using game-based tech to rethink safety, with sensor-based VR applications now used toward just such ends [Khan et al., 2021]. Another example is SwarmCity, [Garcia-Aunon, et al., 2019], a predictive model of traffic patterns, pedestrians, climate, and pollution built in Unity [2005] that, when combined with swarms of quadcopters, can yield algorithms for managing traffic and vehicular response to it. As cars themselves become "smart," game engines become tools for training them, as with Tesla AI researchers training cars through simulated San Francisco streets in models developed in Unreal [Lambert, 2022]. As with digital twins, game-based technologies contribute to the automotive industry by enabling the design and modeling of complex dynamics that can be deployed to platforms then used to facilitate understanding across multiple stakeholder groups, including consumers themselves.

3.3.13 Disaster Preparation/Response (36)

Similarly, game-based platforms have been used to model emergency and disaster scenarios, coordinate communication across individuals and groups, train professionals and the public, and measure readiness. Real-time rendering tools enable teams to model disastrous events, such as mud landslides, so that participants can learn from historical cases and prepare for new ones [Alene et al., 2023]. In Building Information Modeling (BIM), teams model foot traffic flows during fire evacuations in large buildings [Wang, et al., 2014]. The low-cost and user-friendly nature of game-based tools enabled teams to engage in multi-site, distributed planning activities at relatively low cost [Mol, et al., 2008]. New AR/VR approaches, such as the Active Shooter Response Training (ASRT) train users to respond to dangerous situations in schools [Harden, 2019]. Evacuation training through such methods has improved students' time in making it to safety [Huang et al., 2023]. Finally, game-based tools such as Enhanced Dynamic Geo-Social Environment (EDGE), train first responders to deal with terrorist attacks [Dubno, 2017]. A primary strength of such tools is their capacity for virtual deployment so that professionals from multiple agencies (local, county, and state police, fire departments, and hospitals) can coordinate activities from the individual to the agency level [Zhou et al., 2016]. In many respects, games enable virtual, distributed versions of traditional fire drills but with more complex scenarios, higher fidelity simulated activities, more extensive contingency efforts, opportunities to coordinate across levels of public services, and with better opportunities to learn from feedback on performance through data gathered during training events.

3.3.14 Art & Other Entertainment Domains (24)

Game-based innovations are increasingly used for prototyping, visualization, real-time rendering, and creative expression in film, television, theater, visual arts, and fashion. Set designers and directors use game engines to prototype scenes during pre-production, leveraging integrated physics, camera, animation, and lighting packages to rapidly produce scenes. For example, the television series' His Dark Materials [Thorne, 2019] and Black Mirror [Brooker, 2011] were prototyped with game technologies to weigh final production decisions. Theater professionals have also used such tools to design sets, scenes, characters, and costumes [Ruecker et al., 2013]. Fashion designers have begun prototyping virtual clothing catwalks, suggesting opportunities to also broaden participation in such events beyond those physically present [Hong & Ge, 2022]. Game-based innovations have also made their way into live broadcasts, with ESPN using AR data and immersive 3D real-time rendering to create interactive visualizations for analysis [Lemire, 2022]. In are and other entertainment domains, the adoption of game-based tech follows a pattern from initial prototyping to integration in live events, with futures in real-time virtual participation in live events including sports, fashion, and music, theater, and interactive entertainment.

3.3.15 Artificial Intelligence (22)

Since the advent of computing, Games have been used as a context for conceptualizing, theorizing, developing, testing, training, and validated artificial intelligence, from classic games such as Chess [Heath & Allum, 1997] to Go to digital ones such as Pong, Diplomacy and Starcraft [Prince, 2020]. Researchers have long created algorithms for computers to play chess [e.g., Shannon, 1950] in an effort to test an AI system's mettle against humans [cf. Turing, 1953]. Game engines provide researchers novel contexts for considering collaborative, cooperative, competitive, and deceptive behaviors, providing contexts that are relatively cheap, accessible, and testable [den Heijer & Goede, 2014]. Contemporary research with games and AI employs "self-play" in which artificial intelligence plays against itself in order to evolve its algorithms, with researchers developing and integrating new wrinkles (constraints or challenges) such as "embedded critics" [Liu et al., 2021]. Real-time strategy games such as Starcraft [1998] are a fruitful platform for research as they enable AI researchers to explore tactical and strategic decision-making, plan recognition, and learning in a context that bridges academics and industry [Robertson & Watson, 2014]. Diplomacy [Calhamer, 1959], a game with deep social negotiation mechanics, has been used to train game language-based AI [Bakhtin et al., 2022]. Future directions involve applying such game-based AI systems to new tasks, such as identifying satellite parts so that spaceships can better conduct tasks including, rendezvous, docking, maintenance, and refueling [Zhao et al., 2023]. Research in this domain highlights the sweet spot that games-based research can occupy between basic and applied research, wherein researchers can tackle genuine problems with novel methods in ways that gathers data to test their efficacy, which, when combined with human actors in systems, suggests new and innovative models for future research and design.

3.3.16 Cybersecurity (21)

Cybersecurity has leveraged games' capacity to engage users, sustain interest, integrate practice with progressive difficulty, situate learning within narratives, and engage the player's identity [VanSteenburg, 2017]. One research study identified 181 games related to cybersecurity across platforms [Coendraad, et al., 2020]; for example, the game Phishy [Gokul et al., 2018] in which players attempt to phish their friends in a series of gamified tasks. Research demonstrates the effectiveness of such approaches compared to more traditional instruction when the desired outcomes are behavioral and not just declarative [Jensen, et al., 2022]. Whereas discussions of gamification often situate incentive structures in competition with deeper, more simulated mechanics [Bogost, 2011], cybersecurity research reminds us that well-structured incentive systems are powerful generators of performance improvements [Condly et al., 2003].

4. CONCLUSIONS

Existing reviews and meta-analyses have established a strong empirical foundation for understanding how games support learning and engagement across educational and applied contexts (Clark, Tanner-Smith, & Killingsworth, 2016; Connolly et al., 2012; Wouters et al., 2013). These syntheses, alongside work in gamification research (e.g., Deterding et al., 2011; Hamari, Koivisto, & Sarsa, 2014), largely concentrate on outcomes within formal education or specific intervention types. Our review extends this lineage by mapping where and how game innovations—technological, methodological, and conceptual—diffuse across sectors beyond education, revealing patterns of impact that encompass but are not limited to instructional domains. In doing so, it complements and broadens the long-standing Simulation & Gaming and ISAGA traditions (e.g., Crookall, 2011; Wilkinson, 2016), which have for decades documented games and simulations as tools for learning, training, and organizational change. By integrating those insights with contemporary evidence on cross-sector diffusion, this review situates the study of games as part of a larger ecology of innovation, pointing to ways that design practices and technologies migrate between entertainment and non-entertainment spheres. Building on this broader context, the following sections summarize key themes that emerge from this specific contribution to the research base.

4.1 Real-time interactive simulations for multiple stakeholders

Cutting across all domains reviewed, a key affordance of games is their capacity to dynamically represent complex systems in ways that multiple stakeholder groups can engage with. Complex systems can be represented through many forms, with tools such as Maya [1998] enabling designers to animate environmental, physical, human, or even synthetic systems and their combinations. Likewise, professional communities, such as medical researchers or automotive designers, have specialized tools well-suited to their practices. Game-based innovations add unique value to existing domains by enabling different constituencies—most notably, the public—to engage with such representations. Digital twin projects such as the Vancouver airport [Redins, 2022] excel when shared representations are crucial to these disciplinary communities. In health, the 3D nature of games is thus far less important than the generation of player data; here, games are the context in which player data can be gathered. Urban planning games enable different constituencies to engage with urban systems. Automotive companies employ game engines so that designers, consumers, and marketers can learn about one another's needs. Scientific discovery games like Fold.it [2008] allow natural scientists, data scientists, and educators to explore new horizons in the study of complex phenomena. Assessment tools such as CyGames [Reese, 2015] enable students, parents, teachers, and administrators to examine student/player data for inferences about learning. Game-based collaborative tools like SCORE [Boy, 2016] allow constituents to collectively consider the design of nuclear power plants. The most compelling example for thinking through this capacity may be digital twins, an approach that could easily be employed by domains beyond architecture, engineering, and construction to model potential policy decisions, novel creative expressions, and even new domains of inquiry. One could imagine, for example, human digital twins that enable medical professionals, patients, or caregivers to examine the impact of treatments and regimens before they are prescribed.

4.2 Engagement

Games are often designed to be engaging, and thus, not surprisingly, frequently used to make other activities more endurable or appealing. Such uses include math games, exergames, games for environmental science, games as marketing, tourism games, and others. Indeed, nearly every domain surveyed included examples in which games were used to increase user interest or motivation to participate in particular activities. At the same time, research consistently reminds us that engagement is not automatic: effective design requires thoughtful alignment between game mechanics, player motivation, and context of use. Evidence for games' capacity to pique interest in a topic is the predominance of reports on game-based innovations in the domain of education [Steinkuehler & Squire, 2024]. Research on exergaming reaches similar conclusions but also notes that exercise habits often fail to endure beyond the intervention (even when interest in exercise may persist). The most compelling reports of games' impact through engagement may be with aging and ailing populations, where engagement with and in the world is a goal in and of itself; here, the effects of game-based technologies can be remarkable, improving cognitive and emotional health in dramatic and enduring ways. Still, these successes highlight not only the motivational power of games but also the challenge of designing experiences that sustain engagement over time. Indeed, reading such reports of game element integration into treatments for cancer or cystic fibrosis patients only underscores the importance of agency and hope—and the role of play in both.

4.3 Data and Game Analytics

Analyses of player data can improve the quality of interventions (e.g. level design in a math game), evidence the efficacy of an intervention (e.g., compliance in an exercise game), and inform our fundamental understandings of phenomena (e.g., knowledge of collaborative patterns in a protein-folding game). Researchers, practitioners, and policymakers have been particularly hopeful for games' capacity to close loops between design, research, and policy. On the one hand, evidence suggests that games can provide data leading to improvements in practice in areas as diverse as emergency preparedness and automotive design. On the other hand, as writing this review 15 years after the release of Farmville [2009], a game famously built on extensive A/B testing, surely highlights, the impact of game data on practice seems limited. In education, multiple examples exist of the use of game data for assessing learning [e.g., Delacruz et al., 2010; Reese, 2015]. At the same time, game-based assessments are not regularly integrated into the daily practices of most teachers or students in any diagnostic way. Examining uses of game and player data across sectors, successes are most evident in cases where there's strong alignment between the purpose of playing the game, organizational intentions, game play models, and the face validity of the data gathered. In emergency preparedness, for example, reducing time to evacuate a building is a direct, measurable outcome closely tied to organizational goals and the purpose of the intervention. In contrast, educational interventions have most often used data to detect secondary or indicator variables only, such as time on task, failure, or implicit learning, which organizations may not value eagerly [Rowe et al., 2015]. This review suggests that game data can improve practices, generate inferences about player behavior, and enable program evaluation, but there needs to be a close alignment among organizational goals, data collected, and game play for this to hold true.

4.4 Gamification

Games are also used to restructure experiences in a much broader sense, such as in higher education where gamification structures (organizing content around challenges and rewards rather than content delivery along pre-determined timelines) are more common. Gamification has been criticized for being a canned, one-size-fits all solution based on behaviorist models rather than nuanced design solutions to problems. This review suggests instead that gamification approaches can be good metaphors for structuring experience in powerful ways. In domains such as computer science, in which information can be delivered and assessed digitally, gamification can offer a model for personalizing learner trajectories across content within the well-defined domain. Gamification strategies have also proven successful in exercise, health care, and marketing—although research also questions whether behaviors persist after the end of such programs.

4.5 Games "In the Wild" and "In the Laboratory"

In each domain surveyed, there is a relationship arises between a game's succeed "in the wild" of commercial entertainment and researchers' and practitioners' interest in its application within their domain. In cases where commercial games or their elements are transferred and applied within novel fields, the game often already instantiates particular theories of interest or pushes the field in new or interesting ways. As with the impact of Sim City [Wright, 1989], a game now over 30 years old as of this writing, the true impact of specific games may not be evident until the Minecraft [Persson, 2011] generation enters adulthood. Urban planning's long relationship with Sim City is a prime example of such delayed estimates of impact as is health's long relationship with exergaming via Dance, Dance Revolution [1998] or Pokemon Go [2016]. Thus, one might anticipate more educational spin-offs from Minecraft [Persson, 2011] given its instantiation of constructivist and sociocultural theories of learning, but it may well be too early to tell—or perhaps the emergence of a wholly new relationship on the market, one in which the game's publisher actually promotes and incorporates its educational use rather than fears it (believing, as the old adage goes, that labeling a game "educational" meant the death of its marketability to players). This common sequence of games "in the wild," domain professional interest, and emerging bespoke games within the domain suggests a roadmap for those seeking to make games for impact.

4.6 Impact of Games on Other Sectors

This review reveals thousands of reports describing game-based innovations making an impact in domains beyond entertainment, ranging from large-scale adoptions in fields like education, healthcare, and the military as well as in more narrowly focused transfer domains such as urban planning, architecture, and cultural tourism. At the same time, there were fewer reports of games having widespread, integrated impact than one might imagine. While reports were highest in the field of education, no game has permeated classrooms in the past twenty years any more than Oregon Trail [Rawitsch et al., 1971], originally developed by three teachers as part of a student-teaching project and later distributed widely through the Minnesota Educational Computing Consortium (Miller, 2024). Similarly, game-based approaches are also used in health but not yet a part of standard practice in many crucial areas. Indeed, the closest thing one might find to thorough integration may be the military, where games and game-based tech are used broadly in training and simulation. This review also uncovers relative siloing in each sector, with cross-sector integration and knowledge-sharing rare. Looking across domains, a sweet spot appears to be in game platforms enabling the rapid development of human-scale environments (such as cityscapes) that allow constituents to explore interesting choices within them; when disciplinary designers want users to have agency, games and game-based tools prove their usefulness.

5 FUTURE DIRECTIONS

With the commercial industry more than 50 years old and the generation who first grew up with them approaching retirement age, video games are indeed making meaningful inroads in domains beyond entertainment. Looking across sectors reveals that a benefit of games and technology may be less the technologies themselves and more a playful orientation they exemplify in overcoming challenge. Games for health is again a useful case in point, with a playful orientation to health associated with physical, emotional, and mental well-being. Game-based approaches to urban planning or even emergency preparedness invite users to join in authoring solutions not just inheriting them. To be sure, game-related technologies are made for play and thus offer affordances useful in its accomplishment, such as real-time control of avatars or communicative feedback from complex systems. Indeed, many of the most effective cases uncovered herein are cases in which designers leverage games to invite broad constituents—from power plant engineer teams to automotive designers to research communities—to gather around a set of common problems wherein the game functions as an object to be collectively thought with and through and about. These two cross-domain themes about the power of play orientation and of models as tools for collaborative cognition, offer future designers both caveats to constrain design and lanterns held high to light the path.

While this review takes a deliberately conservative approach—restricting inclusion to documented cases across scholarly, trade, and governmental sources—the resulting corpus necessarily reflects the emphases of academic publication, with impacts most visible in humanistic and educational fields. At the same time, other forms of influence remain less systematically represented. For example, early video games and game technologies were pivotal in driving advancements in graphics hardware, from the early development of commodity chipsets (e.g., Voodoo, nVidia) that displaced proprietary workstation architectures (e.g., SGI, Sun) to later innovations such as per-vertex and per-pixel pipelines that enabled large-scale parallelization in GPU computation. These technological pathways helped catalyze subsequent breakthroughs in areas such as artificial intelligence and real-time visualization. Such infrastructure-level contributions—where game design and engineering have spurred advances in computing more broadly—fall outside the scope of the present analysis but clearly warrant dedicated investigation in future research.

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REFERENCES

Abt, C. C. (1970). Serious games: The art and science of games that simulate life. New York: Viking.

Abu-Hammad, R.M. & Hamtini, T.M. (2023). A gamification approach for making online education as effective as in-person education in learning programming concepts. International Journal of Emerging Technologies in Learning 18, 28–49. https://doi.org/10.3991/ijet.v18i07.37175

Akram A.K., M., Hafizudddin M., M., Zulfadli N., M., Mahmud, A. Manaf, R.A. (2022). A systematic review on the effectiveness of computer games application in patient rehabilitation. International Journal of Public Health & Clinical Sciences, 9, 14–29.

Alencar, N.E.S., Pinto, M.A.O., Leite, N.T., de Silva, C.M.V. (2022). Serious games for sex education of adolescents and youth: Integrative literature review. Cien Saude Colet 27, 3129–3138. https://doi.org/10.1590/1413-81232022278.00632022

Alene, G. H., Vicari, H., Irshad, S., Perkis, A., Bruland, O., & Thakur, V. (2023). Realistic visualization of debris flow type landslides through virtual reality. Landslides, 20(1), 13–23. https://doi.org/10.1007/s10346-022-01948-x

Aljafari, A., ElKarmi, R., Nasser, O., Atef, A., & Hosey, M. T. (2022). A video-game-based oral health intervention in primary schools—A randomised controlled trial. Dentistry Journal, 10(5), Article 5. https://doi.org/10.3390/dj10050090

Alves, A. & Soares, A.M. (2013). Evaluating the use of avatars in e-commerce. Portuguese Journal of Marketing, 31, 37–52.

America's Army [video game]. (2002). United States Army.

Andersson, J. (2013). Europa Universalis IV. Paradox Interactive.

Anderson, E.F., McLoughlin, L., Liarokapis, F., Peters, C., Petridis, P., De Freitas, S. (2010). Developing serious games for cultural heritage: A state-of-the-art review. Virtual Reality 14, 255–275. https://doi.org/10.1007/s10055-010-0177-3

Animal Crossing [video game]. (2001). Nintendo.

Arpin, R. (2022, April 22). Get out. Association for Talent Development. https://www.td.org/magazines/td-magazine/get-out

Asave, V., Ionita, V.-V., Moldoveanu, F., & Moldoveamu, A. (2009). 3D real-time parallel volume rendering using nvidia cuda for medical imaging. Annals of DAAAM & Proceedings, 1483–1484.

Assassin's Creed [Video game]. (2007). Ubisoft.

Astle, A. & Writer, S. (2022, December 8). CALM research finds gaming has a positive impact on mental health. Pocketgamer.biz. https://www.pocketgamer.biz/data-and-research/80392/calm-research-finds-gaming-has-a-positive-impact-on-mental-health/

"Automotive smart cockpit design trend report 2022: 3D Unreal Engine presents lucrative opportunities for innovation." (2022). Business Wire. https://www.businesswire.com/news/home/20221216005311/en/Automotive-Smart-Cockpit-Design-Trend-Report-2022-3D-Unreal-Engine-Presents-Lucrative-Opportunities-for-Innovation---ResearchAndMarkets.com

Bağ, H. & Çalık, M. (2021). Designing an argumentation-based educational digital game to teach the subject of force. Physics Education, 56, 035002. https://doi.org/10.1088/1361-6552/abdae8

Bai, S., Hew, K.F., Gonda, D.E., Huang, B., & Liang, X. (2022). Incorporating fantasy into gamification promotes student learning and quality of online interaction. International Journal of Educational Technology in Higher Education 19, 1–26. https://doi.org/10.1186/s41239-022-00335-9

Bailenson, J. (2020, September 18). Is VR the future of corporate training? Harvard Business Review. https://hbr.org/2020/09/is-vr-the-future-of-corporate-training

Bakhtin, A., Brown, N., Dinan, E., Farina, G., Flaherty, C., Fried, D., Goff, A., Gray, J., Hu, H., Jacob, A.P., Komeili, M., Konath, K., Kwon, M., Lerer, A., Lewis, M., Miller, A.H., Mitts, S., Renduchintala, A., Roller, S. & Rowe, D. (2022). Human-level play in the game of Diplomacy by combining language models with strategic reasoning. Science 378, 1067–1074.

Balela, M.S. & Mundy, D. (2015). Analysing cultural heritage and its representation in video games. In Proceedings of the DiGRA 2015 Conference. https://dl.digra.org/index.php/dl/article/view/721

Bardelli, S., Del Corso, G., Ciantelli, M., Del Pistoia, M., Lorenzoni, F., Fossati, N., Scaramuzzo, R.T. & Cuttano, A. (2022). Improving pediatric/neonatology residents' newborn resuscitation skills with a digital serious game: DIANA. Frontiers in Pediatrics 10, 842302. https://doi.org/10.3389/fped.2022.842302

Bauer, A., Flatten, J. & Popovic, Z. (2017). Analysis of problem-solving behavior in open-ended scientific-discovery game challenges. Paper presented at the 10th International Conference on Educational Data Mining (EDM), Wuhan, China, Jun 25-28.

Bavelier, B. & Green, C.S. (2016, July 1). Brain tune-up from action video game play. Scientific American. https://www.scientificamerican.com/article/brain-tune-up-from-action-video-game-play/

Bediou, B., Adams, D. M., Mayer, R. E., Tipton, E., Green, C. S., & Bavelier, D. (2018). Meta-analysis of action video game impact on perceptual, attentional, and cognitive skills. Psychological Bulletin, 144(1), 77–110. https://doi.org/10.1037/bul0000130

Beecham, M. (2016, September 15). Why game companies are driving automotive design – Unreal Engine Q&A. Just-Auto.com. https://www.just-auto.com/interview/why-games-technology-is-driving-automotive-design-unreal-engine-qa/

Bell-Gawne, K., Stenerson, M., Shapiro, B. & Squire, K. (2013). Meaningful play: The intersection of video games and environmental policy. World Future Review, 5, 244–250. https://doi.org/10.1177/1946756713497472

Berger, J., Bawab, N., De Mooij, J., Sutter Widmer, D., Szilas, N., De Vriese, C. & Bugnon, O. (2018). An open randomized controlled study comparing an online text-based scenario and a serious game by Belgian and Swiss pharmacy students. Currents in Pharmacy Teaching and Learning 10, 267–276. https://doi.org/10.1016/j.cptl.2017.11.002

Berger, W. & Staley, P. (2014). Assassin's Creed III: The complete unofficial guide, a teacher's limited edition. Well Played, 3(1), 1-10.

Berkovsky, S., Freyne, J. & Coombe, M. (2012). Physical activity motivating games: Be active and get your own reward. ACM Transactions on Computer-Human Interaction (TOCHI) 19, 1–41.

Bethea, D.P., Castillo, D.C. & Harvison, N. (2014). Use of simulation in occupational therapy education: Way of the future? The American Journal of Occupational Therapy, 68(Supplement_2), S32–S39. https://doi.org/10.5014/ajot.2014.012716

Bitrián A.P., Catalán G.S. (2022). Boosting brand engagement and loyalty through gamified loyalty programmes. UCJC Business & Society Review 19, 144–181.

Bleakley, C.M., Charles, D., Porter-Armstrong, A., McNeill, M.D.J., McDonough, S.M. & McCormack, B. (2015). Gaming for health: a systematic review of the physical and cognitive effects of interactive computer games in older adults. Journal of Applied Gerontology, 34, NP166–NP189. https://doi.org/10.1177/0733464812470747

Bonnechère, B., Jansen, B., Omelina, L., & Van Sint Jan, S. (2016). The use of commercial video games in rehabilitation: A systematic review. International Journal of Rehabilitation Research 39, 277–290.

Bossen, D., Broekema, A., Visser, B., Brons, A., Timmerman, A., van Etten-Jamaludin, F., Braam, K. & Engelbert, R. (2020). Effectiveness of serious games to increase physical activity in children with a chronic disease: Systematic review with meta-analysis. Journal of Medical Internet Research, 22, e14549. https://doi.org/10.2196/14549

Bourdillon, A.T., Garg, A., Wang, H., Woo, Y.J., Pavone, M. & Boyd, J. (2023). Integration of reinforcement learning in a virtual robotic surgical simulation. Surgical Innovation 30, 94–102. https://doi.org/10.1177/15533506221095298

Bodner, M., & Coulson, A. (2021). Randomized trial of elementary school ST Math software intervention reveals significant efficacy. In Grantee Submission. https://eric.ed.gov/?id=ED616922

Bogost, I. (2011, August 9). Gamification is bullshit. The Atlantic. https://www.theatlantic.com/technology/archive/2011/08/gamification-is-bullshit/243338/

Bowman, E. (2023, February 19). A future NBA app feature lets fans virtually replace a player in a live game. Georgia Public Broadcasting. https://www.gpb.org/news/2023/02/19/future-nba-app-feature-lets-fans-virtually-replace-player-in-live-game

Boy, G.A., Jani, G., Manera, A., Memmott, M., Petrovic, B., Rayad, Y., Stephane, L. & Suri, N. (2016). Improving collaborative work and project management in a nuclear power plant design team: A human-centered design approach. Annals of Nuclear Energy 94, 555–565. https://doi.org/10.1016/j.anucene.2015.12.039

Breien, F.S. & Wasson, B. (2021). Narrative categorization in digital game-based learning: Engagement, motivation & learning. British Journal of Educational Technology 52, 91–111.

Brooker, C. (2011). Black mirror [Television series]. Zeppotron.

Bryceson, K.P., Leigh, S., Sarwar, S. & Grøndahl, L. (2022). Affluent effluent: Visualizing the invisible during the development of an algal bloom using systems dynamics modelling and augmented reality technology. Environmental Modelling & Software 147, 105253. https://doi.org/10.1016/j.envsoft.2021.105253

Bulik, B.S. (2009). What do video games have to do with marketing? I mean, we're not all selling to teen boys. Advertising Age, 80, 54–54.

Butler, O., Herr, K., Willmund, G., Gallinat, J., Kühn, S. & Zimmermann, P. (2020). Trauma, treatment and Tetris: video gaming increases hippocampal volume in male patients with combat-related posttraumatic stress disorder. Journal of Psychiatry & Neuroscience, 45, 279–287. https://doi.org/10.1503/jpn.190027

Button, K. (2009). Video game, with real people, coaches U.S. soldiers on cultures. Defense News 24, 54–56.

Cai, Y., Chiew, R., Nay, Z.T., Indhumathi, C. & Huang, L. (2017). Design and development of VR learning environments for children with ASD. Interactive Learning Environments 25, 1098–1109.

Calhamer, A.B. (1959). Diplomacy. Wizards of the Coast.

Carolan, J. & Zielezinski (2019, May 21). Debunking the 'gold standard' myths in edtech efficacy. EdSurge. https://www.edsurge.com/news/2019-05-21-debunking-the-gold-standard-myths-in-edtech-efficacy

Caton, H. & Greenhill, D. (2013). The effects of gamification on student attendance and team performance in a third-year undergraduate game production module. European Conference on Games Based Learning, (pp. 88-96). Reading: Academic Conferences International Limited.

Chalhoub, M., Khazzaka, A., Sarkis, R. & Sleiman, Z. (2018). The role of smartphone game applications in improving laparoscopic skills. Advances in medical education and practice 9, 541.

Chan, C.-S., Chan, Y. & Fong, T.H.A. (2020). Game-based e-learning for urban tourism education through an online scenario game. International Research in Geographical & Environmental Education 29, 283–300. https://doi.org/10.1080/10382046.2019.1698834

Chang, C. -J., Chang, M. -H., Liu, C. -C., Chiu, B. -C., Fan Chiang, S. -H., Wen, C. -T., Hwang, F. -K., Chao, P. -Y., Chen, Y. -L., & Chai, C. -S. (2017). An analysis of collaborative problem-solving activities mediated by individual-based and collaborative computer simulations. Journal of Computer Assisted Learning, 33(6), 649–662. https://doi.org/10.1111/jcal.12208

Charlier, N., Zupancic, N., Fieuws, S., Denhaerynck, K., Zaman, B., & Moons, P. (2016). Serious games for improving knowledge and self-management in young people with chronic conditions: A systematic review and meta-analysis. Journal of the American Medical Informatics Association, 23, 230–239. https://doi.org/10.1093/jamia/ocv100

Chatham, R.E. (2007). Games for Training. Communications of the ACM 50, 36–43. https://doi.org/10.1145/1272516.1272537

Che, T., Peng, Y., Zhou, Q., Dickey, A. & Lai, F. (2023). The impacts of gamification designs on consumer purchase: A use and gratification theory perspective. Electronic Commerce Research & Applications 59, N.PAG-N.PAG. https://doi.org/10.1016/j.elerap.2023.101268

Clark, D.B., Hernández-Zavaleta, J.E. & Becker, S. (2023). Academically meaningful play: Designing digital games for the classroom to support meaningful gameplay, meaningful learning, and meaningful access. Computers & Education 194, 104704. https://doi.org/10.1016/j.compedu.2022.104704

Clark, D.B., Tanner-Smith, E.E. & Killingsworth, S.S. (2016). Digital games, design, and learning: A systematic review and meta-analysis. Review of Educational Research 86, 79–122. https://doi.org/10.3102/0034654315582065

Cloude, E., Carpenter, D., Dever, D.A., Lester, J. & Azevedo, R. (2021). Game-based learning analytics for supporting adolescents' reflection. Journal of Learning Analytics 8, 51–72. https://doi.org/10.18608/jla.2021.7371

Coenraad, M., Pellicone, A., Ketelhut, D.J., Cukier, M., Plane, J. & Weintrop, D. (2020). Experiencing cybersecurity one game at a time: A systematic review of cybersecurity digital games. Simulation & Gaming 51, 586–611. https://doi.org/10.1177/1046878120933312

Colder Carras, M., Kalbarczyk, A., Wells, K., Banks, J., Kowert, R., Gillespie, C. & Latkin, C. (2018). Connection, meaning, and distraction: A qualitative study of video game play and mental health recovery in veterans treated for mental and/or behavioral health problems. Social Science & Medicine 216, 124–132. https://doi.org/10.1016/j.socscimed.2018.08.044

Comello, M.L.G., Francis, D.B., Marshall, L.H. & Puglia, D.R. (2016). Cancer survivors who play recreational computer games: Motivations for playing and associations with beneficial psychological outcomes. Games for Health Journal 5, 286–292. https://doi.org/10.1089/g4h.2016.0003

Condly, S.J., Clark, R.E. & Stolovitch, H.D. (2003). The effects of incentives on workplace performance: A meta-analytic review of research studies. Performance Improvement Quarterly 16, 46–63.

https://doi.org/10.1111/j.1937-8327.2003.tb00287.x

Condor, C. (1994). Rex Ronan: Experimental Surgeon. Raya Systems.

Connolly, T. M., Boyle, E. A., MacArthur, E., Hainey, T., & Boyle, J.M. (2012). A systematic literature review of empirical evidence on computer games and serious games. Computers & Education, 59(2), 661-686. https://doi.org/10.1016/j.compedu.2012.03.004.

"Why Unreal Engine is a 'transformative' tool for advertising." (n.d.). Contagious. Retrieved May 17, 2023, from https://www.contagious.com/news-and-views/how-unreal-engine-can-transform-advertising

Cooper, L.R. (2018). Digital game-based learning and the mathematics achievement of gifted students. Unpublished Dissertation Thesis. Liberty University.

CPI Office. (2021, February 24). Cultural Heritage Center develops traditional Potawatomi games into online experiences. Potawatomi.Org. https://www.potawatomi.org/blog/2021/02/24/cultural-heritage-center-develops-traditional-potawatomi-games-into-online-experiences/

Crookall, D. (2011). Serious Games, Debriefing, and Simulation/Gaming as a Discipline. Simulation & Gaming, 41(6), 898-920. https://doi.org/10.1177/1046878110390784 (Original work published 2010)

Crosse, J. (2021, November 1). Gaming tech aiding car industry. Autocar. https://www.autocar.co.uk/car-news/business-tech%2C-development-and-manufacturing/how-gaming-technology-aiding-car-industry

D'Agostini, M.M., Aredes, N.D.A., Campbell, S.H. & Fonseca, L.M.M. (2020). Serious game e-baby família: An educational technology for premature infant care. Brazilian Journal of Nursing, 73, e20190116. https://doi.org/10.1590/0034-7167-2019-0116

Dance, Dance, Revolution [video game]. (1998). Konami, Bemani.

DARWARS Ambush! [video game]. (2004). Total Immersion Software.

Das, R., Keep, B., Washington, P. & Riedel-Kruse, I.H. (2019). Scientific discovery games for biomedical research. Annual Review of Biomedical Data Science, 2, 253–279. https://doi.org/10.1146/annurev-biodatasci-072018-021139

Davidson, J.G. & Eckert, R.K. (1983). Math Blaster. Davidson & Associates.

Davis, M., Shilling, R., Mayberry, A., McCree, J., Bossant, P. Dossett, S., Buhl, C., Chang, C., Champin, E., Wiglesworth, T, & Zyda, M. (2003). Making America's Army. In B. Laurel (Ed.), Design Research: Methods and Perspectives. Cambridge MA: MIT Press.

De Giglio, L., De Luca, F., Prosperini, L., Borriello, G., Bianchi, V., Pantano, P., & Pozzilli, C. (2015). A low-cost cognitive rehabilitation with a commercial video game improves sustained attention and executive functions in multiple sclerosis: A pilot study. Neurorehabilitation & Neural Repair, 29(5), 453–461. CINAHL Complete. https://doi.org/10.1177/1545968314554623

Delacruz, G. C., Chung, G. K. W. K., & Baker, E. L. (2010). Validity evidence for games as assessment environments (CRESST Report 773). Los Angeles: University of California, Los Angeles, National Center for Research on Evaluation, Standards, and Student Testing (CRESST).

de-Marcos, L., Garcia-Lopez, E. & Garcia-Cabot, A. (2016). On the effectiveness of game-like and social approaches in learning: Comparing educational gaming, gamification & social networking. Computers & Education 95, 99–113. https://doi.org/10.1016/j.compedu.2015.12.008

den Heijer, & M., Goede, R. (2014). Implementing an intelligent agent in a known, observable, discrete and deterministic environment using a scriptable game engine. European Conference on Data Mining 2014 and International Conferences Intelligent Systems and Agents 2014 and Theory and Practice in Modern Computing 2014 – Part of the Multi Conference on Computer Science and Information Systems, MCCSIS 2014.

Deterding, S., Dixon, D., Khaled, R., & Nacke, L. E. (2011). From game design elements to gamefulness: Defining "gamification." Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments (pp. 9–15). ACM. https://doi.org/10.1145/2181037.2181040

Doker, M.F. & Kirkangicoglu, C. (2018). Promotion of cultural heritages through a virtual museum platform: Case study Hagia Sophia. Sakarya University Journal of Science, 22, 97–105.

Drummond, D., Monnier, D., Tesnière, A. & Hadchouel, A. (2017). A systematic review of serious games in asthma education. Pediatric Allergy and Immunology, 28, 257–265. https://doi.org/10.1111/pai.12690

Du, W., Zhong, X., Jia, Y., Jiang, R., Yang, H., Ye, Z., & Zong, Z. (2022). A novel scenario-based, mixed-reality platform for training nontechnical skills of battlefield first aid: Prospective interventional study. JMIR Serious Games 10, 246–256. https://doi.org/10.2196/40727

Dubno, D., 2017. Terrorist Training for First Responders. Popular Mechanics, 20–20.

Dunbar, N. E., Jensen, M. L., Miller, C. H., Bessarabova, E., Lee, Y.-H., Wilson, S. N., Elizondo, J., Adame, B. J., Valacich, J., Straub, S., Burgoon, J. K., Lane, B., Piercy, C. W., Wilson, D., King, S., Vincent, C., & Schuetzler, R. M. (2017). Mitigation of Cognitive Bias with a Serious Game: Two Experiments Testing Feedback Timing and Source. International Journal of Game-Based Learning, 7(4), 86–100.

ECS adds canine interventions to enhance medical training scenarios (2023, January 31). Halldale Group. (2023). https://www.halldale.com/articles/20722-mst-ecs-adds-canine-interventions-to-enhance-medical-training-scenarios

Ee, R.W.X., Yap, K.Z. & Yap, K.Y.-L. (2018). Herbopolis - A mobile serious game to educate players on herbal medicines. Complementary Therapies in Medicine, 39, 68–79. https://doi.org/10.1016/j.ctim.2018.05.004

Eiben, C.B., Siegel, J.B., Bale, J.B., Cooper, S., Khatib, F., Shen, B.W., Foldit Players, Stoddard, B.L., Popovic, Z., & Baker, D., (2012). Increased Diels-Alderase activity through backbone remodeling guided by Foldit players. Nature Biotechnology 30, 190–192. https://doi.org/10.1038/nbt.2109

Entertainment Software Association (2023). 2023 Essential facts about the US video games industry. https://www.theesa.com/wp-content/uploads/2023/07/ESA2023EssentialFactsFINAL07092023.pdf

Ersen, Z. B., & Ergül, E. (2022). Trends of Game-Based Learning in Mathematics Education: A Systematic Review. International Journal of Contemporary Educational Research, 9(3), 603–623.

Esteve-González, V., Camacho, M., Gisbert-Cervera, M. & Vicens, J. (2016). Teachers as game designers through storytelling. Proceedings of the European Conference on Games Based Learning 1, 973–976.

Euklidiadas, M.M. (2022, March 30). Cities and videogames: an unexpectedly constructive relationship. Tomorrow.City. https://tomorrow.city/a/city-planning-games

Farmville [Video game]. (2009). Zynga.

Fish M.T., Russoniello C.V., & O'Brien K. (2018). Zombies vs anxiety: An augmentation study of prescribed video game play compared to medication in reducing anxiety symptoms. Simulation Gaming, 49(5), 553–66. doi: 10.1177/1046878118773126.

Fold.It [Video game]. (2008). University of Washington.

Future Market Insights. (2023). Serious Game Market. https://www.futuremarketinsights.com/reports/serious-game-market

Garcia-Aunon, P., Roldán, J.J. & Barrientos, A. (2019). Monitoring traffic in future cities with aerial swarms: Developing and optimizing a behavior-based surveillance algorithm. Cognitive Systems Research 54, 273–286. https://doi.org/10.1016/j.cogsys.2018.10.031

Garcia-Fernandez, J. & Medeiros, L. (2019). Cultural heritage and communication through simulation videogames—A validation of Minecraft. Heritage 2, 2262–2274. https://doi.org/10.3390/heritage2030138

Garrett, R., Smith, T., Griffin, M., & Yisak, M. (2020). A randomized field study of a teacher professional development program using mixed-reality simulation to develop instructional practice. In Society for Research on Educational Effectiveness. https://eric.ed.gov/?id=ED621545

Gaydos, M.J. & Squire, K.D. (2012). Role playing games for scientific citizenship. Cultural Studies of Science Education 7, 821–844. https://doi.org/10.1007/s11422-012-9414-2

Gee, J.P. (2004). What video games have to teach us about learning and literacy. New York: Palgrave Macmillan.

Giannotti, D., Patrizi, G., Rocco, G. D., Vestri, A. R., Semproni, C. P., Fiengo, L., Pontone, S., Palazzini, G., & Redler, A. (2013). Play to become a surgeon: Impact of Nintendo WII training on laparoscopic skills. PLOS ONE, 8(2), e57372. https://doi.org/10.1371/journal.pone.0057372

Glaser, B.G. & Strauss, A.L. (1967) The discovery of grounded theory: Strategies for qualitative research. Chicago: Aldine.

Gokul, CJ, Pandit, S., Vaddepalli, S., Tupsamudre, H., Banahatti, V. & Lodha, S. (2018). PHISHY - A serious game to train enterprise users on phishing awareness. In Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts, CHI PLAY '18 Association for Computing Machinery (pp. 169–181) New York, NY. https://doi.org/10.1145/3270316.3273042

Gómez-Prada, U. & Gómez-Sandoval, O. (2019). SAMI: Serious videogame of bovine cattle farms in Unity supported in System Dynamics: Revista UIS Ingenierías 18, 15–22. https://doi.org/10.18273/revuin.v18n4-2019001

Gonzales-Chávez, M.A. & Vila-Lopez, N. (2021). Designing the best avatar to reach millennials: Gender differences in a restaurant choice. Industrial Management & Data Systems 121, 1216–1236. https://doi.org/10.1108/IMDS-03-2020-0156

Goodwin, J. (2016). Discovering nature through mobile gaming. Doctoral Dissertation. University of Otago.

Gordon, E. & Manosevitch, E. (2011). Augmented deliberation: Merging physical and virtual interaction to engage communities in urban planning. New Media & Society 13, 75–95.

Gredler, M. E. (1996). Educational Games and Simulations: A Technology in Search of a (Research) Paradigm. Technology, 39, 521-540.

Gordon, A.K. (1970). Games for Growth: Educational Games in the Classroom. Science Research Associates. ISBN: 9780574173867, 0574173862

Grübel, J., Thrash, T., Aguilar, L., Gath-Morad, M., Chatain, J., Sumner, R.W., Hölscher, C. & Schinazi, V.R. (2022). The Hitchhiker's Guide to fused twins: A review of access to digital twins in situ in smart cities. Remote Sensing 14, 3095. https://doi.org/10.3390/rs14133095

Gue, S., Ray, J. & Ganti, L. (2022). Gamification of graduate medical education in an emergency medicine residency program. International Journal of Emergency Medicine 15, 1–7.

Gutierrez, B., Kaatz, A., Chu, S., Ramirez, D., Samson-Samuel, C. & Carnes, M. (2014). "Fair Play": A videogame designed to address implicit race bias through active perspective taking. Games Health, 3, 371–378. https://doi.org/10.1089/g4h.2013.0071

Halbrook, Y.J., O'Donnell, A.T. & Msetfi, R.M. (2019). When and how video games can be good: A review of the positive effects of video games on well-being. Perspectives on Psychological Science 14, 1096–1104. https://doi.org/10.1177/1745691619863807

Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does gamification work?—A literature review of empirical studies on gamification. Proceedings of the 47th Hawaii International Conference on System Sciences (HICSS) (pp. 3025–3034). IEEE. https://doi.org/10.1109/HICSS.2014.377

Hammond, J., Jones, V., Hill, E.L., Green, D. & Male, I. (2014). An investigation of the impact of regular use of the Wii Fit to improve motor and psychosocial outcomes in children with movement difficulties: A pilot study. Child: Care, Health & Development 40, 165–175.

Hannig, A., Kuth, N., Özman, M., Jonas, S., & Spreckelsen, C. (2012). eMedOffice: A web-based collaborative serious game for teaching optimal design of a medical practice. BMC Medical Education, 12, 104. https://doi.org/10.1186/1472-6920-12-104

Harden, Security, (2019). New mixed reality active shooter response training. PR Newswire US.

Hayes, D. & Odell, J. (2020). Change: A Homeless Survival Experience. Delve Interactive.

Hayward, B. (2017). The gamification of customer behaviours. Credit Control 38, 36–40.

Hazel, J., Kim, H.M. & Every-Palmer, S. (2022). Exploring the possible mental health and wellbeing benefits of video games for adult players: A cross-sectional study. Australasian Psychiatry 30, 541–546. https://doi.org/10.1177/10398562221103081

Heath, D. & Allum, D. (1997). The historical development of computer chess and its impact on artificial intelligence. AAAI Workshop Papers. https://aaai.org/papers/063-ws97-04-013/

Heffner, J. L., Watson, N. L., Serfozo, E., Kelly, M. M., Reilly, E. D., Kim, D., Baker, K., Scout, N. F. N., & Karekla, M. (2021). An avatar-led digital smoking cessation program for sexual and gender minority young adults: Intervention development and results of a single-arm pilot trial. JMIR Formative Research, 5(7), e30241. https://doi.org/10.2196/30241

Helfand, C. (2017, September 6). Pfizer puts hemophilia spin on ultrapopular Minecraft to educate young patients Fierce Pharma. https://www.fiercepharma.com/marketing/pfizer-puts-hemophilia-spin-ultrapopular-minecraft-to-educate-young-patients

Herodotou, C., Kambouri, M., & Winters, N. (2014). Dispelling the myth of the socio-emotionally dissatisfied gamer. Computers in Human Behavior, 32, 23–31. https://doi.org/10.1016/j.chb.2013.10.054

Higher Education Video Game Alliance (2015). Priming the Pump 2015: Survey of Program Graduates. Washington DC: HEVGA.

Hong, Y. & Ge, Y., (2022). Design and analysis of clothing catwalks taking into account Unity's immersive virtual reality in an artificial intelligence environment. Computational Intelligence and Neuroscience, 2861767–12. https://doi.org/10.1155/2022/2861767

How AR and VR are transforming training in manufacturing. (2018, August 9). Control Engineering. https://www.controleng.com/articles/how-ar-and-vr-are-transforming-training-in-manufacturing/

"How the video game 'Stray' is helping rescue real-life cats." (2023). PBS NewsHour. https://www.pbs.org/newshour/show/how-the-video-game-stray-is-helping-rescue-real-life-cats

Hsieh, H.-C. (2019). Use of a gaming platform for balance training after a stroke: A randomized trial. Archives of Physical Medicine and Rehabilitation 100, 591–597. https://doi.org/10.1016/j.apmr.2018.11.001

Hu, Y.-H., Yu, H.-Y., Tzeng, J.-W. & Zhong, K.-C. (2023). Using an avatar-based digital collaboration platform to foster ethical education for university students. Computers & Education 196, N.PAG-N.PAG.

Huang, Y., Guo, Z., Chu, H. & Sengupta, R. (2023). Evacuation simulation implemented by ABM-BIM of Unity in students' dormitory based on delay time. ISPRS International Journal of Geo-Information 12, 160-. https://doi.org/10.3390/ijgi12040160

Huang, Y.-M., Huang, S.-H. & Wu, T.-T. (2014). Embedding diagnostic mechanisms in a digital game for learning mathematics. Educational Technology Research & Development 62, 187–207. https://doi.org/10.1007/s11423-013-9315-4

Hulick, K. (2017). Video games level up life skills: Playing games in a group can strengthen abilities such as communication and resourcefulness. Science News for Students 1–1.

Insinna, V. (2013). Contracts highlight growing role of video game training. National Defense 97, 24–25.

Itō, M. (2009). Engineering play: A cultural history of children's software. Cambridge MA: MIT Press. ISBN 978-0-262-29155-2.

Jantakun, T., Jantakun, K. & Jantakoon, T. (2023). Systematic review of smart classroom for hard skills training in augmented and virtual reality environments. Education Quarterly Reviews, 6(1), 280-294. DOI: 10.31014/aior.1993.06.01.706

Jelsma, L. D., Cavalcante Neto, J. L., Smits-Engelsman, B., Targino Gomes Draghi, T., Araújo Rohr, L., & Tudella, E. (2023). Type of active video-games training does not impact the effect on balance and agility in children with and without developmental coordination disorder: A randomized comparator-controlled trial. Applied Neuropsychology: Child, 12(1), 64–73. https://doi.org/10.1080/21622965.2022.2030740

Jensen, M.L., Wright, R.T., Durcikova, A., Karumbaiah, S. (2022). Improving phishing reporting using security gamification. Journal of Management Information Systems 39, 793–823. https://doi.org/10.1080/07421222.2022.2096551

Kaczmarek, L. D., Misiak, M., Behnke, M., Dziekan, M., & Guzik, P. (2017). The Pikachu effect: Social and health gaming motivations lead to greater benefits of Pokémon GO use. Computers in Human Behavior, 75, 356–363.

Kataria, D.I. (2022). Positive trend between video game warm-up and surgical simulation performance. Medical Dialogues. https://medicaldialogues.in/ophthalmology/news/positive-trend-between-video-game-warm-up-and-surgical-simulation-performance-104145

Kaya, O.S. & Ercag, E. (2023). The impact of applying challenge-based gamification program on students' learning outcomes: Academic achievement, motivation and flow. Education and Information Technologies,1–26. https://doi.org/10.1007/s10639-023-11585-z

Khaldi, A., Bouzidi, R. & Nader, F. (2023). Gamification of e-learning in higher education: A systematic literature review. Smart Learning Environments 10.

Khan, M., Magee, L., Pollio, A. & Salazar, J.F. (2021). Counter-fun, scholarly legitimacy, and environmental engagement – or why academics should code games. First Monday, 26 (2-1). https://doi.org/10.5210/fm.v26i2.11427

Khan, N., Muhammad, K., Hussain, T., Nasir, M., Munsif, M., Imran, A.S. & Sajjad, M. (2021). An adaptive game-based learning strategy for children road safety education and practice in virtual space. Sensors (14248220) 21, 3661–3661. https://doi.org/10.3390/s21113661

Kim, J. & Castelli, D.M. (2021). Effects of gamification on behavioral change in education: A meta-analysis.

International Journal of Environmental Research and Public Health, 18. https://doi.org/10.3390/ijerph18073550

Kim, Y.J., Knowles, M.A., Scianna, J., Lin, G. & Ruipérez-Valiente, J.A. (2023). Learning analytics application to examine validity and generalizability of game-based assessment for spatial reasoning. British Journal of Educational Technology 54, 355–372.

Kinect. (2010). Microsoft.

Klabbers, J.H.G. (2009). The Saga of ISAGA. Simulation & Gaming 40(1), 30–47. https://doi.org/10.1177/1046878107310604

Kodu (2009). Microsoft's FUSE Labs

Kolson, K. (1996). The politics of SimCity. Political Science and Politics 29, 43–46. https://doi.org/10.2307/420191

Kwok, R. (2019). Can climate change games boost public understanding? Proceedings of the National Academy of Sciences, 116, 7602–7604. https://doi.org/10.1073/pnas.1903508116

Lambert, F. (2022, September 20). Tesla is creating a simulation of San Francisco in Unreal Engine Electrek. https://electrek.co/2022/09/20/tesla-creating-simulation-san-francisco-unreal-engine/

Lagro, J., van de Pol, M. H. J., Laan, A., Huijbregts-Verheyden, F. J., Fluit, L. C. R., & Olde Rikkert, M. G. M. (2014). A randomized controlled trial on teaching geriatric medical decision making and cost consciousness with the serious game GeriatriX. Journal of the American Medical Directors Association, 15(12), 957.e1-6. https://doi.org/10.1016/j.jamda.2014.04.011

Langston, J. (2017, April 24). Scientific discovery game significantly speeds up neuroscience research process. University of Washington News. https://www.washington.edu/news/2017/04/24/scientific-discovery-game-significantly-speeds-up-neuroscience-research-process/

LaPensee, E. (2019). When Rivers Were Trails Indian Land Tenure Foundation.

Lawler, J. (2015). Earth Games. University of Washington.

Lee, C.-C. & Hussain, J. (2022). Optimal behavior of environmental regulations to reduce carbon emissions: A simulation-based dual green gaming model. Environmental Science and Pollution Research, 29, 56037–56054. https://doi.org/10.1007/s11356-022-19710-0

Lee, H., Parsons, D., Kwon, G., Kim, J., Petrova, K., Jeong, E. & Ryu, H. (2016). Cooperation begins: Encouraging critical thinking skills through cooperative reciprocity using a mobile learning game. Computers & Education 97, 97–115. https://doi.org/10.1016/j.compedu.2016.03.006

Lee, L.-A., Wang, S.-L., Chao, Y.-P., Tsai, M.-S., Hsin, L.-J., Kang, C.-J., Fu, C.-H., Chao, W.-C., Huang, C.-G., Li, H.-Y. & Chuang, C.-K. (2018). Mobile technology in e-learning for undergraduate medical education on emergent otorhinolaryngology-head and neck surgery disorders: Pilot randomized controlled trial. JMIR Medical Education 4, e8. https://doi.org/10.2196/mededu.9237

Lee, M.J.W., Eustace, K., Fellows, G., Bytheway, A. & Irving, L. (2005). Rochester Castle MMORPG: Instructional gaming and collaborative learning at a Western Australian school. Australasian Journal of Educational Technology, 21. https://doi.org/10.14742/ajet.1314

Leggett, D. (2022, June 8). Volvo cars and Epic Games collaborate in HMI. Just Auto. https://www.just-auto.com/news/volvo-cars-and-epic-games-collaborate-in-hmi/

Lemire, J. (2022, October 3). How ESPN's Monday night football incorporates real-time AR data visualization from DMED technology to enhance viewing experience for fans. Sports Business Journal. https://www.sportsbusinessjournal.com/Daily/Issues/2022/10/03/Technology/espn-monday-night-football-augmented-reality-data-visualization-technology.aspx

Leonard, J., Buss, A., Gamboa, R., Mitchell, M., Fashola, O. S., Hubert, T., & Almughyirah, S. (2016). Using robotics and game design to enhance children's self-efficacy, STEM attitudes, and computational thinking skills. Journal of Science Education and Technology, 25(6), 860–876.

Lewis, J.E., Trojovsky, M. & Jameson, M.M. (2021). New social horizons: Anxiety, isolation, and Animal Crossing during the COVID-19 pandemic. Frontiers in Virtual Reality 2, 14.

Li Y, Liu Y, Ye L, Sun J, & Zhang J. (2021). Pokémon GO! GO! GO! The impact of Pokémon GO on physical activity and related health outcomes. mHealth, 7(51). doi:10.21037/mhealth-20-121

Liao, X., Zhao, X., Wang, Z., Han, K., Tiwari, P., Barth, M.J. & Wu, G. (2022). Game theory-based ramp merging for mixed traffic with Unity-SUMO co-simulation. IEEE Transactions on Systems, Man and Cybernetics:

Systems, 52, 5746–5757. https://doi.org/10.1109/TSMC.2021.3131431

Lieberman, D.A. (1997). Interactive video games for health promotion: Effects on knowledge, self-efficacy, social support, and health. Health Promotion and Interactive Technology: Theoretical Applications and Future Directions,103-120.

Lieberman D.A. (2001). Management of chronic pediatric diseases with interactive health games: Theory and research findings. The Journal of Ambulatory Care Management, 24(1), 26-38. DOI: 10.1097/00004479-200101000-00004.

Lin, J., Spraragen, M., Blythe, J., & Zyda, M. (2011). EmoCog : Computational integration of emotion and cognitive architecture. In Proceedings of the Twenty-Fourth International Florida Artificial Intelligence Research Society Conference (pp.111-116). https://cdn.aaai.org/ocs/2625/2625-11120-1-PB.pdf

Lindberg, R.S.N., Laine, T.H. & Haaranen, L. (2019). Gamifying programming education in K-12: A review of programming curricula in seven countries and programming games. British Journal of Educational Technology 50, 1979–1995.

Linden, A. & Fenn, J. (2003, May 30). Understanding Gartner's hype cycles. Gartner. Archived from the original on 2023-06-27. Retrieved 2023-06-27.

Liu, M. & Curet, M. (2015). A review of training research and virtual reality simulators for the da Vinci Surgical System. Teaching and Learning in Medicine 27, 12–26. https://doi.org/10.1080/10401334.2014.979181

Liu, S., Cao, J., Wang, Y., Chen, W. & Liu, Y. (2021). Self-play reinforcement learning with comprehensive critic in computer games. Neurocomputing 449, 207– 213.

Liu, W., Zeng, N., McDonough, D.J. & Gao, Z. (2020). Effect of active video games on healthy children's fundamental motor skills and physical fitness: A systematic review. International Journal of Environmental Research and Public Health, 17. https://doi.org/10.3390/ijerph17218264

Loerzel, V.W., Clochesy, J.M. & Geddie, P.I. (2020). Using serious games to increase prevention and self-management of chemotherapy-induced nausea and vomiting in older adults with cancer. Oncology Nursing Forum, 47, 567–576. https://doi.org/10.1188/20.ONF.567-576

Loban, R. (2021). Modding Europa Universalis IV: An informal gaming practice transposed into a formal learning setting. E-Learning and Digital Media 18, 530–556. https://doi.org/10.1177/20427530211022964

Lohse, K. R., Boyd, L. A. & Hodges, N. J. (2016). Engaging environments enhance motor skill learning in a computer gaming task. Journal of Motor Behavior, 48(2), 172–182.

Lu, S., Fang, C. & Xiao, X. (2023). Virtual scene construction of wetlands: A case study of Poyang Lake, China. ISPRS International Journal of Geo-Information 12, 49. https://doi.org/10.3390/ijgi12020049

Lv, Z., Tek, A., Silva, F.D., Empereur-mot, C., Chavent, M. & Baaden, M. (2013). Game on, science - How video game technology may help biologists tackle visualization challenges. PLOS ONE 8, e57990. https://doi.org/10.1371/journal.pone.0057990

Ma, B.D., Ng, S.L., Schwanen, T., Zacharias, J., Zhou, M., Kawachi, I. & Sun, G. (2018). Pokémon GO and physical activity in Asia: Multilevel study. Journal of Medical Internet Research 20, e9670. https://doi.org/10.2196/jmir.9670

Manriquez, G.B. (1995). Bronckie the Bronchiasaurus. Raya Systems.

Marino, M.T., Basham, J.D. & Beecher, C.C. (2011). Using video games as an alternative science assessment for students with disabilities and at-risk learners. Science Scope 34, 36–40.

Markets and Markets (2021). Game-based learning market - Size, growth, report & analysis. https://www.marketsandmarkets.com/Market-Reports/game-based-learning-market-146337112.html

Marlatt, R. (2018). Get in the game: Promoting justice through a digitized literature study. Multicultural Perspectives 20, 222–228.

Martin, W., Silander, M. & Rutter, S. (2019). Digital games as sources for science analogies: Learning about energy through play. Computers & Education 130, 1–12.

Mason, R. & Turner, L. (2018). Serious gaming: A tool to educate health care providers about domestic violence. Health Care for Women International 39, 859–871.

Matzko, R.O., Mierla, L. & Konur, S. (2023). Novel ground-up 3D multicellular simulators for synthetic biology CAD integrating stochastic Gillespie simulations benchmarked with topologically variable SBML models.

Genes 14, 154. https://doi.org/10.3390/genes14010154

Maya. [Graphics application]. (1998). Autodesk, Inc.

Mayer, R. E. (2019). Computer games in education. Annual Review of Psychology, 70, 531–549. https://doi.org/10.1146/annurev-psych-010418-102744

McCallum, S. (2012). Gamification and serious games for personalized health. Studies in Health Technology and Informatics, 177, 85–96.

McDaniel, R. & Fanfarelli, J. (2016). Building better digital badges. Simulation & Gaming 47, 73–102. https://doi.org/10.1177/1046878115627138

McGaffey, M., Linden, A.Z., Sears, W., Monteith, G., Khosa, D.K. & Blois, S.L. (2023). Teaching accuracy through repeated gamified echography training (TARGET): Assessment of an ultrasound skill simulator in teaching ultrasound-guided needle placement, a prospective observational study. Journal of Veterinary Medical Education e20220131. https://doi.org/10.3138/jvme-2022-0131

McWilliams, L. (2004). Full Spectrum Warrior. THQ.

Mead, C. (2013). War play: Video games and the future of armed conflict. Publishers Weekly 260, 158–158.

Meyer, (2014, July 8) The effect of the ‘new’ e-learning on soft skills training. TD Magazine. https://www.td.org/content/td-magazine/the-effect-of-the-new-e-learning-on-soft-skills-training?__queryID=6cc1158d993a464afd0bf2fb84728ea5&objectID=7rsbsRE3B7H6zz5FJgkEs6&__position=1&index= atd_composable_prod_en-US

Merchant-Borna, K., Jones, C.M.C., Janigro, M., Wasserman, E.B., Clark, R.A., & Bazarian, J.J. (2017). Evaluation of Nintendo Wii balance board as a tool for measuring postural stability after sport-related concussion. Journal of Athletic Training 52, 245–255. https://doi.org/10.4085/1062-6050-52.1.13

Miao, F., Kozlenkova, I. V., Wang, H., Xie, T., & Palmatier, R. W. (2022). An emerging theory of avatar marketing. Journal of Marketing, 86(1), 67–90. Business Source Complete. https://doi.org/10.1177/0022242921996646

Miller, A. (2024, December 19). The Oregon Trail: How a 50-year-old video game defined America BBC..https://www.bbc.com/future/article/20241219-the-oregon-trail-how-a-50-year-old-video-game-defined-america

Minnery, J. & Searle, G. (2014). Toying with the city? Using the computer game SimCityTM4 in planning education. Planning Practice & Research 29, 41–55.

MIND Research Institute (1998). ST Math. [Video game].

Mól, A.C.A., Jorge, C.A.F. & Couto, P.M. (2008). Using a game engine for VR simulations in evacuation planning. IEEE Computer Graphics & Applications 28, 6–12.

Montagni, I., Mabchour, I. & Tzourio, C. (2020). Digital gamification to enhance vaccine knowledge and uptake: Scoping review. JMIR Serious Games 8, e16983. https://doi.org/10.2196/16983

Morii, J. (2006). Wii Sports. Nintendo EAD.

Mubin, O., Alnajjar, F., Al Mahmud, A., Jishtu, N. & Alsinglawi, B. (2022). Exploring serious games for stroke rehabilitation: A scoping review. Disability & Rehabilitation: Assistive Technology 17, 159–165. https://doi.org/10.1080/17483107.2020.1768309

Nagoshi, T. (2001). Super Monkey Ball. Sega.

Nakano, C.M., Moen, E., Byun, H.S., Ma, H., Newman, B., McDowell, A., Wei, T. & El-Naggar, M.Y. (2016). iBET: Immersive visualization of biological electron-transfer dynamics. Journal of Molecular Graphics & Modelling 65, 94–99. https://doi.org/10.1016/j.jmgm.2016.02.009

Nardo, A. & Gaydos, M. (2021). ‘Wicked problems’ as catalysts for learning in educational ethics games. Ethics and Education 16, 492–509. https://doi.org/10.1080/17449642.2021.1979283

Nastis, S.A. & Pagoni, E.G. (2019). Gamification of global climate change: an experimental analysis. Studies in Nonlinear Dynamics & Econometrics 23, N.PAG-N.PAG. https://doi.org/10.1515/snde-2017-0105

Ni, M. Y., Hui, R. W. H., Li, T. K., Tam, A. H. M., Choy, L. L. Y., Ma, K. K. W., Cheung, F., & Leung, G. M. (2019). Augmented reality games as a new class of physical activity interventions? The impact of Pokémon Go Uue and gaming intensity on physical activity. Games for Health Journal, 8(1), 1–6. https://doi.org/10.1089/g4h.2017.0181

Nimrod, G. (2011). The fun culture in seniors’ online communities. The Gerontologist, 51(2), Article 2.

Nintendo Wii. (2006). Nintendo.

National Research Council. (2011). Learning science through computer games and simulations. Washington, DC: The National Academies Press. https://doi.org/10.17226/13078.

Ohman, A.W., Hasan, N. & Dinulescu, D.M. (2014). Advances in tumor screening, imaging, and avatar technologies for high-grade serous ovarian cancer. Frontiers in Oncology 4, 322. https://doi.org/10.3389/fonc.2014.00322

Packy and Marlon (1995). Raya Systems.

Pando Cerra, P., Fernández Álvarez, H., Busto Parra, B., & Iglesias Cordera, P. (2022). Effects of using game-based learning to improve the academic performance and motivation in engineering studies. Journal of Educational Computing Research, 60(7), 1663–1687. https://doi.org/10.1177/07356331221074022

Parisod, H., Pakarinen, A., Axelin, A., Danielsson-Ojala, R., Smed, J. & Salanterä, S. (2017). Designing a health-game intervention supporting health literacy and a tobacco-free life in early adolescence. Games for Health Journal 6, 187–199. https://doi.org/10.1089/g4h.2016.0107

Parrish, K. (2005). Treat lazy eye the fun way. Better Homes & Gardens, 83(5), 280–280.

Patel, M.S., Small, D.S., Harrison, J.D., Hilbert, V., Fortunato, M.P., Oon, A.L., Rareshide, C.A.L. & Volpp, K.G. (2021). Effect of behaviorally designed gamification with social incentives on lifestyle modification among adults with uncontrolled diabetes: A randomized clinical trial. JAMA Network Open 4, e2110255–e2110255. https://doi.org/10.1001/jamanetworkopen.2021.10255

Persson, M. (2011). Minecraft. Mojang Studios.

Peterson, S. (n.d.). A serious look at game-based learning. EdSurge News. Retrieved May 16, 2023, from https://www-edsurge-com.cdn.ampproject.org/c/s/www.edsurge.com/amp/news/2022-10-25-a-serious-look-at-game-based-learning

Pingmuang, P. & Koraneekij, P. (2022). Mobile-assisted language learning using task-based approach and gamification for enhancing writing skills in EFL students. Electronic Journal of e-Learning 20, 623–638.

Peng, W., Crouse, J.C. & Lin, J.-H. (2013). Using active video games for physical activity promotion: A systematic review of the current state of research. Health Education & Behavior 40, 171–192.

Peng, W., Lin, J.-H. & Crouse, J. (2011). Is playing exergames really exercising? A meta-analysis of energy expenditure in active video games. Cyberpsychology, Behavior and Social Networking 14, 681–688. https://doi.org/10.1089/cyber.2010.0578

Pham, T.N., Wong, J.N., Terken, T., Gibran, N.S., Carrougher, G.J. & Bunnell, A. (2018). Feasibility of a Kinect ® -based rehabilitation strategy after burn injury. Burns: Journal of the International Society for Burn Injuries 44, 2080–2086. https://doi.org/10.1016/j.burns.2018.08.032

Piot, M., Dechartres, A., Attoe, C., Romeo, M., Jollant, F., Billon, G., Cross, S., Lemogne, C., Layat Burn, C., Michelet, D., Guerrier, G., Tesniere, A., Rethans, J., & Falissard, B. (2022). Effectiveness of simulation in psychiatry for nursing students, nurses and nurse practitioners: A systematic review and meta-analysis. Journal of Advanced Nursing 78, 332–347. https://doi.org/10.1111/jan.14986

Pokémon [Video game]. (1996). Nintendo.

Pokémon Go [Video game]. (2016). Niantic.

Ponticorvo, M., Di Fuccio, R., Di Ferdinando, A., & Miglino, O. (2017). An agent-based modelling approach to build up educational digital games for kindergarten and primary schools. Expert Systems, 34(4), n/a-N.PAG. Academic Search Complete.

Pope, L. (2013). Papers, Please. 3909 LLC.

Pour Rahimian, F., Seyedzadeh, S., Oliver, S., Rodriguez, S. & Dawood, N. (2020). On-demand monitoring of construction projects through a game-like hybrid application of BIM and machine learning. Automation in Construction 110, N.PAG-N.PAG. https://doi.org/10.1016/j.autcon.2019.103012

Poy, R., García, M., Aguilar, A.G. & Arezes, F.C.P. (2019). Wizards, elves and orcs going to high school: How role-playing video games can improve academic performance through visual learning techniques. Education for Information 35, 305–318.

Primack, B.A., Carroll, M.V., McNamara, M., Klem, M.L., King, B., Rich, M., Chan, C.W. & Nayak, S. (2012). Role of video games in improving health-related outcomes: a systematic review. American Journal of

Preventive Medicine 42, 630–638. https://doi.org/10.1016/j.amepre.2012.02.023

Prince, C. (2020, June 12). Video games are the fruit fly of AI research TheGamer. https://www.thegamer.com/video-games-fruit-fly-ai-research/

Project Horseshoe (2017). Coziness in games: An Exploration of safety, softness, and satisfied needs. https://www.projecthorseshoe.com/reports/featured/ph17r3.htm

Puolitaival, T., Sieppi, M., Pyky, R., Enwald, H., Korpelainen, R., & Nurkkala, M. (2020). Health behaviours associated with video gaming in adolescent men: a cross-sectional population-based MOPO study. BMC Public Health 20, 415 (2020). https://doi.org/10.1186/s12889-020-08522-x

Qian, M. & Clark, K.R. (2016). Game-based learning and 21st century skills: A review of recent research. Computers in Human Behavior 63, 50–58. https://doi.org/10.1016/j.chb.2016.05.023

Quiles, C., & Verdoux, H. (2023). Benefits of video games for people with schizophrenia: A literature review. Current Opinion in Psychiatry, 36(3), 184–193. https://doi.org/10.1097/YCO.0000000000000867

Rahaman, H., Johnston, M. & Champion, E. (2021). Audio-augmented arboreality: wildflowers and language. Digital Creativity 32, 22–37. https://doi.org/10.1080/14626268.2020.1868536

Ramani, G.B., Daubert, E.N., Lin, G.C., Kamarsu, S., Wodzinski, A. & Jaeggi, S.M. (2020). Racing dragons and remembering aliens: Benefits of playing number and working memory games on kindergartners' numerical knowledge. Developmental Science 23, e12908. https://doi.org/10.1111/desc.12908

Rantala, A., Vuorinen, A.-L., Koivisto, J., Similä, H., Helve, O., Lahdenne, P., Pikkarainen, M., Haljas, K., & Pölkki, T. (2022). A gamified mobile health intervention for children in day surgery care: Protocol for a randomized controlled trial. Nursing Open, 9(2), 1465–1476. https://doi.org/10.1002/nop2.1143

Rawitsch, D., Heinemann, B., & Dillenberger, P. (1971). Oregon trail [Video game]. Broderbund.

Redins, L. (2022, October 19). Using digital twins to improve operations at Vancouver's airport. Edge Industry Review. https://www.edgeir.com/using-digital-twins-to-improve-operations-at-vancouvers-airport-20221019

Reer, F. & Krämer, N.C. (2018). Psychological need satisfaction and well-being in first-person shooter clans: Investigating underlying factors. Computers in Human Behavior, 84, 383–391. https://doi.org/10.1016/j.chb.2018.03.010

Reese, D.D. (2015). CyGaMEs Selene player log dataset: Gameplay assessment, flow dimensions and non-gameplay assessments. British Journal of Educational Technology 46, 1005–1014.

Re:Mission [Video game]. (2006). HopeLab.

Re-Mission 2: Nanobot's Revenge [Video game]. (2013). HopeLab.

Rhu, J., Lim, S., Kang, D., Cho, J., Lee, H., Choi, G.-S., Kim, J.M. & Joh, J.-W. (2022). Virtual reality education program including three-dimensional individualized liver model and education videos: A pilot case report in a patient with hepatocellular carcinoma. Annals of Hepato-Biliary-Pancreatic Surgery, 26, 285–288. https://doi.org/10.14701/ahbps.21-163

Rice, A. (2022, May 17). VR exercise games could offer hope for delaying dementia. CNET. https://www.cnet.com/health/medical/vr-exercise-games-offer-hope-for-delaying-dementia/

Rixon, D. & Skipton, M.D. (2018). Encouraging information search in accounting cases by using avatars as sources - But students still wanted to be given the information on paper! Journal of Higher Education Theory & Practice 18, 88–101.

Robertson, G. & Watson, I. (2014). A Review of real-time strategy game AI. AI Magazine 35, 75–104. https://doi.org/10.1609/aimag.v35i4.2478

Rodgers, P. (1992). Captain Novolin. Raya Systems.

Rosser J.C., Gentile D.A., Hanigan K. & Danner O.K. (2012). The effect of video game "warm-up" on performance of laparoscopic surgery tasks. Journal of the Society of Laparoscopic & Robotic Surgeons, 16(1), 3-9. DOI: 10.4293/108680812X13291597715664.

Rowe, E., Baker, R.S. & Asbell-Clarke, J. (2015). Strategic game moves mediate implicit science learning, International Educational Data Mining Society. Paper presented at the 8th International Conference on Educational Data Mining (EDM), Madrid, Spain, June 26-29.

Ruecker, S., Grotkowski, A., Gabriele, S., Roberts-Smith, J., Sinclair, S., Dobson, T., Akong, A., Fung, S., Hong, M., DeSouza-Coelho, S. & Rodriguez, O. (2013). Abstraction and realism in the design of avatars for the simulated environment for theatre. Visual Communication 12, 459–472.

Rupp, M.A., Sweetman, R., Sosa, A.E., Smither, J.A. & McConnell, D.S. (2017). Searching for affective and

cognitive restoration: Examining the restorative effects of casual video game play. Human Factors 59, 1096–1107. https://doi.org/10.1177/0018720817715360
Rutherford, T., Kibrick, M., Burchinal, M., Richland, L., Conley, A., Osborne, K., Schneider, S., Duran, L., Coulson, A., Antenore, F., Daniels, A. & Martinez, M.E. (2009). Spatial temporal mathematics at scale: An innovative and fully developed paradigm to boost math achievement among all learners. Institute of Education Science. https://ies.ed.gov/funding/grantsearch/details.asp?ID=830
Schmidt, R., Sauer, S. & Wilkening, J. (2021). Re-building a historical cityscape with virtual reality. Abstracts of the International Cartographic Association, 3, 1–2. https://doi.org/10.5194/ica-abs-3-261-2021
Scratch [Programming language]. (2003). Scratch Foundation.
Schnurr, C. (1999). Rainbox Six: Rogue Spear. NA: Red Storm Entertainment.
Schrier, K. (2016). Knowledge games. Johns Hopkins University Press. https://doi.org/10.1353/book.47461
Scorpio, M., Laffi, R., Masullo, M., Ciampi, G., Rosato, A., Maffei, L. & Sibilio, S. (2020). Virtual reality for smart urban lighting design: Review, applications and opportunities. Energies, 13(15) 3809. https://doi.org/10.3390/en13153809
Sell, K., Lillie, T. & Taylor, J. (2008). Energy expenditure during physically interactive video game playing in male college students with different playing experience. Journal of American College Health 56, 505–512.
Semeraro, F., Frisoli, A., Loconsole, C., Mastronicola, N., Stroppa, F., Ristagno, G., Scapigliati, A., Marchetti, L. & Cerchiari, E. (2017). Kids (learn how to) save lives in the school with the serious game Relive. Resuscitation 27–32. https://doi.org/10.1016/j.resuscitation.2017.04.038
Seo, Y. (2013). Electronic sports: A new marketing landscape of the experience economy. Journal of Marketing Management 29, 1542–1560. https://doi.org/10.1080/0267257X.2013.822906
Shannon, C.E. (1950). Programming a computer for playing chess. Philosophical Magazine, 41(7), 256-275.
Shohieb, S.M., Doenyas, C. & Elhady, A.M. (2022). Dynamic difficulty adjustment technique-based mobile vocabulary learning game for children with autism spectrum disorder. Entertainment Computing 42, 100495. https://doi.org/10.1016/j.entcom.2022.100495
Shonfeld, M. & Greenstein, Y. (2021). Factors promoting the use of virtual worlds in educational settings. British Journal of Educational Technology 52, 214–234.
Shute, V. J., Ventura, M., & Kim, Y. (2013). Assessment and learning of qualitative physics in Newton’s playground. Journal of Educational Research 106, 423–430.
Silverman, D. (2022, April 11). Mind-controlled VR games could aid stroke rehabilitation. Imperial News. Imperial College London. https://www.imperial.ac.uk/news/235375/mind-controlled-vr-games-could-stroke-rehabilitation/
Santos, I. K. D., Medeiros, R. C. da S. C. de, Medeiros, J. A. de, Almeida-Neto, P. F. de, Sena, D. C. S. de, Cobucci, R. N., Oliveira, R. S., Cabral, B. G. de A. T., & Dantas, P. M. S. (2021). Active video games for improving mental health and physical fitness-An alternative for children and adolescents during social isolation: An overview. International Journal of Environmental Research and Public Health, 18(4). https://doi.org/10.3390/ijerph18041641
Sawyer, B., & Rejeski, D. (2002). Serious games: Improving public policy through game-based learning and simulation. Woodrow Wilson International Center for Scholars.
Schreier, J. (2017). Blood, sweat, and pixels: The triumphant, turbulent stories behind how video games are made. New York: Harper.
Sneed, A. (2013, March 6). What SimCity teaches us about real cities of the future. Slate. https://slate.com/technology/2013/03/simcity-2013-what-the-urban-planning-game-tells-us-about-future-cities.html
Stokes, B., Bar, F., Baumann, K., Caldwell, B., Schrock, A. (2021). Urban furniture in digital placemaking: Adapting a storytelling payphone across Los Angeles. Convergence 27, 711–726. https://doi.org/10.1177/1354856521999181
Steinkuehler, C., & Squire, K. (2024). Gaming in educational contexts. In P.A. Schutz & K.R. Muis (Eds.), the APA Handbook of Educational Psychology, 4th edition (pp. 674-695). New York: Routledge. ISBN: 9780429433726
Stewart, J. & Misuraca, G. (2013). The industry and policy context for digital games for empowerment and inclusion: Market analysis, future prospects and key challenges in videogames, serious games and gamification. European Commission Joint Research Centre: Institute for Prospective Technological Studies. https://doi.org/10.2791/88361
S.T. Math [Educational software]. (1998). MIND Research Institute.
Sirály, E., Szabó, Á., Szita, B., Kovács, V., Fodor, Z., Marosi, C., Salacz, P., Hidasi, Z., Maros, V., Hanák, P.,

Csibri, É., & Csukly, G. (2015). Monitoring the early signs of cognitive decline in elderly by computer games: An MRI study. PLoS ONE, 10(2), 1–14.
Sun, L., Chen, X. & Ruokamo, H. (2021). Digital game-based pedagogical activities in primary education: A review of ten years' studies. International Journal of Technology in Teaching and Learning 16, 78–92.
StarCraft: BroodWar [Video game]. (1998). Blizzard Entertainment.
Sweeney, T. (1995). Unreal [Game engine]. Epic Games.
Sydora, C., Lei, Z., Siu, M.F.F., Han, S. & Hermann, U. (2021). Critical lifting simulation of heavy industrial construction in gaming environment. Facilities 39, 113–131. https://doi.org/10.1108/F-08-2019-0088
Tan, A.J.Q., Lee, C.C.S., Lin, P.Y., Cooper, S., Lau, L.S.T., Chua, W.L. & Liaw, S.Y. (2017). Designing and evaluating the effectiveness of a serious game for safe administration of blood transfusion: A randomized controlled trial. Nurse Education Today 55, 38–44. https://doi.org/10.1016/j.nedt.2017.04.027
Pajitnov, A. (1985) Tetris. [Video game].
Then, J.W., Shivdas, S., Yahaya, T.S., Ab Razak, N.I. & Choo, P.T. (2020). Gamification in rehabilitation of metacarpal fracture using cost-effective end-user device: A randomized controlled trial. Journal of Hand Therapy 33, 235–242. https://doi.org/10.1016/j.jht.2020.03.029
Thiagarajan, S., & Stolovitch, H.D. (1978). Instructional simulation games. Englewood Cliffs: Educational Technology.
Thilmany, J. (2005). Games for quick thinking? Mechanical Engineering 127, 16–16.
This War of Mine [Video game]. (2014). 11 Bit Studios.
Thomas, T. H., McLaughlin, M., Hayden, M., Shumaker, E., Trybus, J., Myers, E., Zabiegalski, A., & Cohen, S. M. (2019). Teaching patients with advanced cancer to self-advocate: Development and acceptability of the Strong TogetherTM Serious Game. Games for Health Journal, 8(1), 55–63. https://doi.org/10.1089/g4h.2018.0021
Thorne, J. (2019). His dark materials [Television series]. BBC Studios.
Tokac, U., Novak, E., & Thompson, C. G. (2019). Effects of game-based learning on students' mathematics achievement: A meta-analysis. Journal of Computer Assisted Learning, 35(3), 407–420. https://doi.org/10.1111/jcal.12347
Tomlinson, B., Baumer, E., Yau, M. L. Carpenter, F. L. & Black. R. (2008). A participatory simulation for informal education in restoration ecology. E-Learning, 5(3), 238-255.
Torque [Game engine]. (2009). GarageGames.
Torrado Cespón, M. & Díaz Lage, J.M. (2022). Gamification, online learning and motivation: A quantitative and qualitative analysis in higher education. Contemporary Educational Technology 14(4), Article ep381.
Tough, D., Robinson, J., Gowling, S., Raby, P., Dixon, J. & Harrison, S.L. (2018). The feasibility, acceptability and outcomes of exergaming among individuals with cancer: A systematic review. BMC Cancer 18, 1151. https://doi.org/10.1186/s12885-018-5068-0
Turing, A. M. (1953). Digital computers applied to games, In B.V. Bowden (Ed). Faster than Thought. London: Pitman.
Uluay, G. & Dogan, A. (2020). Pre-service science teachers' learning and teaching experiences with digital games: KODU game lab. Journal of Education in Science, Environment and Health 6(2), 105-119. DOI:10.21891/jeseh.668961
Unity [Game engine]. (2005). Unity Technologies.
U.S. Bureau of Labor Statistics (2023). Software developers, quality assurance analysts, and testers. https://www.bls.gov/ooh/computer-and-information-technology/software-developers.htm
Vagg, T., Tabirca, S., Shortt, C., Fleming, C. & Plant, B. (2019). Improving patient education and the transition process using virtual reality. eLearning & Software for Education 1, 523–526. https://doi.org/10.12753/2066-026X-19-069
Vakili, A., Langdon, R., & Mobini, S. (2016). Cognitive rehabilitation of attention deficits in traumatic brain injury using action video games: A controlled trial. Cogent Psychology, 3(1), N.PAG-N.PAG. https://doi.org/10.1080/23311908.2016.1143732
Vanderstraeten L., Buysschaert, F., De Mulder, V., François, D., Janssens, L., Maes, A., Maes, G., Minnaert, E. & Opdecam, E. (2022). How to welcome first-year students: Best practice of a gamified orientation day. Proceedings of the European Conference on Games Based Learning 588–596.
VanSteenburg, M. (2017). Applications of serious gaming to cybersecurity training and awareness. Dissertation Thesis. Utica College.
Vaterlaus, J.M., Frantz, K., Robecker, T. (2019). "Reliving my childhood dream of being a Pokémon trainer": An exploratory study of college student uses and gratifications related to Pokémon Go. International Journal of Human-Computer Interaction 35, 596–604. https://doi.org/10.1080/10447318.2018.1480911

Velez, J. A. (2015). Extending the theory of bounded generalized reciprocity: An explanation of the social benefits of cooperative video game play. Computers in Human Behavior, 48, 481–491. https://doi.org/10.1016/j.chb.2015.02.015
Waisberg, E., Ong, J., Zaman, N., Kamran, S.A., Lee, A.G. & Tavakkoli, A. (2022). Head-mounted dynamic visual acuity for g-transition effects during interplanetary spaceflight: Technology development and results from an early validation study. Aerospace Medicine and Human Performance 93, 800–805. https://doi.org/10.3357/AMHP.6092.2022
Wang, B., Li, H., Rezgui, Y., Bradley, A. & Ong, H.N. (2014). BIM based virtual environment for fire emergency evacuation. Scientific World Journal, 1, 589016–589016. https://doi.org/10.1155/2014/589016
Wang, H. (2009). Reviving and safeguarding intangible traditional culture: A case study in the use of game technology to revive and safeguard Beijing Opera. Doctoral Thesis. Griffith University.
Wang, K. & Davies, E.G.R. (2015). A water resources simulation gaming model for the Invitational Drought Tournament. Journal of Environmental Management 160, 167–183. https://doi.org/10.1016/j.jenvman.2015.06.007
Wang, X., Guo, C., Yuen, D.A. & Luo, G. (2020). GeoVReality: A computational interactive virtual reality visualization framework and workflow for geophysical research. Physics of the Earth & Planetary Interiors 298, 106312. https://doi.org/10.1016/j.pepi.2019.106312
Wang, Y.-H. (2023). Can gamification assist learning? A study to design and explore the uses of educational music games for adults and young learners. Journal of Educational Computing Research 60, 2015–2035.
Wattanasoontorn, V., Boada, I., García, R. & Sbert, M. (2013). Serious games for health. Entertainment Computing 4, 231–247. https://doi.org/10.1016/j.entcom.2013.09.002
Waweru, B.W., Ng, P.S.J. & Eaw, H.C. (2021). JomGamesy: Using game mechanics to boost intrinsic motivation in school. International Journal of Business Strategy & Automation, 2, 36–52. https://doi.org/10.4018/IJBSA.20210701.oa3
Weatherbed, J. (2022, September 29). Autodesk and Epic Games are joining forces to bring immersion to architecture tools. The Verge. https://www.theverge.com/2022/9/29/23378913/epic-games-autodesk-unreal-engine-partnership-twinmotion-free-design-tool
Wei, L., Zhou, H. & Nahavandi, S. (2019). Haptically enabled simulation system for firearm shooting training. Virtual Reality 23, 217–228. https://doi.org/10.1007/s10055-018-0349-0
Weintrop, D. & Wilensky, U. (2014). Situating programming abstractions in a constructionist video game. Informatics in Education 13, 307–321.
while True: learn(), [Video game]. (2019) Luden.io
Wii Balance Board. [Video game peripheral]. (2007). Nintendo.
Wii Remote. [Video game peripheral]. (2006).Nintendo
Wilkinson, P. (2016). A Brief History of Serious Games. In Dörner, R., Göbel, S., Kickmeier-Rust, M., Masuch, M., Zweig, K. (eds) Entertainment Computing and Serious Games. Lecture Notes in Computer Science, vol 9970. Springer, Cham. https://doi.org/10.1007/978-3-319-46152-6_2
Wouters, P., van Nimwegen, C., van Oostendorp, H., & van der Spek, E.D. (2013). A meta-analysis of the cognitive and motivational effects of serious games. Journal of Educational Psychology, 105(2), 249-265. http://dx.doi.org/10.1037/a0031311
Wright, W. (1989). Sim City. [Video game]. Electronic Arts.
Yang, X. & Delparte, D. (2022). A procedural modeling approach for ecosystem services and geodesign visualization in Old Town Pocatello, Idaho. Land, 11(8), 1228. https://doi.org/10.3390/land11081228
Yen, H.-Y., & Chiu, H.-L. (2021). Virtual reality exergames for improving older adults' cognition and depression: A systematic review and meta-analysis of randomized control trials. Journal of the American Medical Directors Association, 22(5), 995–1002. https://doi.org/10.1016/j.jamda.2021.03.009
Yoshida, A., Kumazaki, H., Muramatsu, T., Yoshikawa, Y., Ishiguro, H. & Mimura, M. (2022). Intervention with a humanoid robot avatar for individuals with social anxiety disorders comorbid with autism spectrum disorders. Asian Journal of Psychiatry 78, 103315. https://doi.org/10.1016/j.ajp.2022.103315
Zhang, F. & Kaufman, D. (2017). Massively multiplayer online role-playing games (MMORPGs) and socio-emotional wellbeing. Computers in Human Behavior 73, 451–458. https://doi.org/10.1016/j.chb.2017.04.008
Zhang, R-Y., Chopin, A., Shibata, K., Lu, Z.-L., Jaeggi, S.M., Buschkuehl, M., Green, C.S., & Bavelier, D. (2021). Action video game play facilitates "learning to learn." Communications Biology, 4(1), 1154. https://doi.org/10.1038/s42003-021-02652-7
Zhao, Y., Zhong, R. & Cui, L. (2023). Intelligent recognition of spacecraft components from photorealistic images based on Unreal Engine 4. Advances in Space Research 71(9), 3761–3774.

https://doi.org/10.1016/j.asr.2022.09.045
Zhou, Z., Chang, J.S.-K., Pan, J. & Whittinghill, D. (2016). Alternate reality game for emergency response training: A review of research. Journal of Interactive Learning Research, 27(1), 77–95.
Zombies, Run! [Video game]. (2012). Six to Start.

APPENDIX: TABLE OF SEED TERMS USED FOR ACADEMIC DATABASE SEARCHES

Seed Term	Role of Games Industry	# Publications
Serious Games		
Education	applied/transferred	9731
Corporate Training & Professional Development	applied/transferred	2403
Marketing & Advertising	applied/transferred	3870
Physical Health & Exergaming	applied/transferred	3376
Military & Defense	applied/transferred	1426
Social Issues & Awareness	applied/transferred	1214
Environment & Sustainability	applied/transferred	1204
Healthcare & Medical Training	applied/transferred	673
Mental Health & Well-Being	applied/transferred	641
Public Policy & Governance	applied/transferred	450
Scientific Research & Data Visualization	applied/transferred	450
Architecture, Engineering & Construction	applied/transferred	308
Cultural Heritage & Tourism	applied/transferred	208
Cybersecurity	applied/transferred	160
Emergency Management & Disaster Response	applied/transferred	26
Cross-Cultural Communication	applied/transferred	22
Urban Planning	applied/transferred	18
Input Devices		
Motion controllers	invented/created	6291
Joysticks	catalyzed/advanced it	584
Gamepads	invented/created	547
Haptic feedback devices	invented/created	94
Game Design Elements		
Avatars	applied/transferred	3057
Rewards	catalyzed/advanced it	1230
Levels & Progression	invented/created	938
Achievements	catalyzed/advanced it	902
Narratives	catalyzed/advanced it	280
Leaderboards	invented/created	165
Quests	catalyzed/advanced it	38
Gamification	invented/created	36
Scarcity	catalyzed/advanced it	14
First Order Impacts of Games		
Physical Health	n/a	3376
Social	n/a	626
Cognition	n/a	508
Wellbeing	n/a	298

Hand-Eye Coordination	n/a	122
Economic	n/a	2
Augmented Reality / Virtual Reality		
VR Platforms		
Oculus Rift/Occulus Quest	invented/created	2069
HTC Vive	invented/created	946
Sony PlayStation VR	invented/created	400
Unity XR	invented/created	172
Meta Quest	invented/created	112
Project Morpheus	invented/created	108
Valve Index VR	invented/created	23
HP Reverb	invented/created	8
AR platforms		
Nintendo Labo VR Kit	invented/created	41
Game Engines	invented/created	920
Unreal Engine	invented/created	367
Unity	invented/created	172
CryEngine	invented/created	32
Godot Engine	invented/created	2
Motion Capture		
Markerless motion capture	catalyzed/advanced it	365
Real-time motion capture	catalyzed/advanced it	47
Facial motion capture	catalyzed/advanced it	29
Physics Engines		
Open Dynamics Engine (ODE)	catalyzed/advanced it	23
Havok	invented/created	14
Bullet Physics	invented/created	13
NVIDIA PhysX	catalyzed/advanced it	10
Chipmunk	invented/created	0
Artificial Intelligence (AI)		
Pathfinding Algorithms		
A* (A star) & its variants	catalyzed/advanced it	0
Dijkstra's algorithm	catalyzed/advanced it	0
Genetic Algorithm	catalyzed/advanced it	0
Decision-Making Algorithms		
Monte Carlo Tree Search (MCTS)	testing ground/petri dish	0
Behavior Trees (BT)	catalyzed/advanced it	0
Finite State Machines (FSMs)	catalyzed/advanced it	0
Utility Systems	catalyzed/advanced it	0
Reinforcement Learning		
Deep Reinforcement Learning	testing ground/petri dish	0
Multi-Agent Reinforcement Learning	catalyzed/advanced it	0
Swarm Intelligence	testing ground/petri dish	0
Search Algorithms	catalyzed/advanced it	0
Minimax	invented/created	0

Alpha-Beta pruning	invented/created	0
Procedural Content Generation (PCG)	catalyzed/advanced it	0
Random Map Generation	catalyzed/advanced it	0
Noise-Based Terrain Generation	catalyzed/advanced it	0
Grammar-Based Generation	catalyzed/advanced it	0
Evolutionary Algorithms	catalyzed/advanced it	0
Generative Adversarial Networks (GANs)	catalyzed/advanced it	0
AI-Driven Game Design (con't)		
Player Modeling	catalyzed/advanced it	0
Narrative Generation	invented/created	0
Dynamic Difficulty Adjustment (DDA)	invented/created	0
Competitions/Events		
General Video Game AI (GVGAI) Competition	testing ground/petri dish	0
Mario AI Championship	testing ground/petri dish	0
StarCraft AI Competition	testing ground/petri dish	0
Fighting Game AI Competition	testing ground/petri dish	0
Hearthstone AI Competition	testing ground/petri dish	0
Graphics & Rendering Techniques		
Sprite-Based Graphics	invented/created	0
Tile-Based Rendering	invented/created	0
Temporal Anti-Aliasing (TAA)	catalyzed/advanced it	0
GPU (Graphics Processing Unit) Development	invented/created	0
Texture Mapping	catalyzed/advanced it	0
Multitexturing	catalyzed/advanced it	0
Bump Mapping	catalyzed/advanced it	0
Normal Mapping	catalyzed/advanced it	0
Global Illumination	catalyzed/advanced it	0
Light Propagation Volumes (LPV)	invented/created	0
Screen Space Ambient Occlusion (SSAO)	invented/created	0
Screen Space Global Illumination (SSGI)	invented/created	0
Radiosity	catalyzed/advanced it	0
Lightmaps	catalyzed/advanced it	0
Real-Time Rendering	catalyzed/advanced it	0
Forward Rendering	catalyzed/advanced it	0
Deferred Rendering	catalyzed/advanced it	0
Real-Time Ray Tracing	catalyzed/advanced it	0
Real-Time Global Illumination	catalyzed/advanced it	0
Hardware-Accelerated Rendering	catalyzed/advanced it	0
Shaders	catalyzed/advanced it	0
Level of Detail (LoD) Techniques	catalyzed/advanced it	0
Post-Processing Effects	catalyzed/advanced it	0
Real-Time Shadows and Reflections	catalyzed/advanced it	0
Occlusion Culling	invented/created	0
Dynamic Lighting	catalyzed/advanced it	0
Real-Time Tessellation	invented/created	0
Screen-Space Subsurface Scattering	catalyzed/advanced it	0

Online & Networked Technologies		
Networking Infrastructures		
Scalable Server Architectures	catalyzed/advanced it	0
Load-Balancing Techniques	catalyzed/advanced it	0
Data Compression Techniques	catalyzed/advanced it	0
Communication Protocols		
User Datagram Protocol (UDP)	catalyzed/advanced it	0
Reliable UDP (RUDP)	catalyzed/advanced it	0
Peer to Peer (P2P) Networking Protocols	catalyzed/advanced it	0
Network Address Translation (NAT) Traversal Protocols	catalyzed/advanced it	0
Routing Management Strategies		
Content Delivery Networks (CDNs)	catalyzed/advanced it	0
Edge Networking	catalyzed/advanced it	0
Traffic Shaping	none/application only	0
Quality of Service (QoS)	catalyzed/advanced it	0
Network Security		
Securing Transaction Protocols	none/application only	0
Authentication Protocols	none/application only	0
Distributed Denial of Service (DDoS) Protection	catalyzed/advanced it	0
Bug Bounty Programs	catalyzed/advanced it	0
Anti-Cheating Mechanisms		
Anti-Cheat Software	invented/created	0
Server-Side Validation	catalyzed/advanced it	0
Hardware ID (HWID)	catalyzed/advanced it	0
Cheat Detection Algorithms	invented/created	0
Machine Learning and AI Approaches		
Anomaly Detection	catalyzed/advanced it	0
Predictive Analysis	catalyzed/advanced it	0
Player Behavior Analysis	catalyzed/advanced it	0
Automated Cheat Classification	catalyzed/advanced it	0
Collaborative Tools & Social Features		
Friends Lists	catalyzed/advanced it	0
Voice Chat (VoIP)	catalyzed/advanced it	0
Guilds & Clans	catalyzed/advanced it	0
Matchmaking & Grouping	invented/created	0
In-game Social Spaces & Virtual Hangouts	invented/created	0
Live Streaming	catalyzed/advanced it	0
Petri Dish (for Primary Research)		
AI	testing ground/petri dish	0
Bodily mechanisms	testing ground/petri dish	0
Cognition & behavior	testing ground/petri dish	0
Social	testing ground/petri dish	0
		51,161